

Toward Fast and Energy-Efficient Access to Cloudlets in Hostile Environments

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Abstract—Cloudlets, which refer to the edge computing services deployed at the proximity of end devices, are key providers of connectivity, storage, and computation resources to many applications. While access to cloudlets is pervasive in typical settings, it can be difficult in challenging, even hostile environments, such as military or post-disaster scenarios, featuring multi-hop communication and energy-constrained end devices. In these cases, cloudlets may have become the only equipment powerful enough to execute life-critical applications, such as battle-field situation awareness, tactic cooperation, and search-and-rescue missions. Quality of these services is greatly influenced by the minimum time that a packet can be delivered, *i.e.*, the *cloudlet access delay* (CAD), whose characteristics remain unknown. To address the open question of fast and efficient cloudlet access, we establish a *packet mobility* model that allows CAD and energy consumption to be analyzed as a function of the initial device-cloudlet distance. We find that the expected CAD scales either linearly or quadratically under distinct types of packet mobility, and the *successful access rate* (SAR) can be bounded by functions of the delay constraint. Based on these findings, we develop a packet shedding algorithm that saves 24% transmission power, and reduces the average CAD by 2%, while maintaining a similar SAR in simulated cloudlet access environments.

Index Terms—Cloud access, mobile ad-hoc network (MANET), performance evaluation.

I. INTRODUCTION

OWING to its flexibility and scalability, cloud computing has been leveraged by numerous applications [1] across the world. However, cloud servers usually reside in server-farms far away from end users (in both geographical and routing senses), resulting in high latencies that are intolerable to delay-sensitive applications, such as augmented reality (AR), language translation, and so on. To bridge the gap between cloud-induced latencies and stringent delay bounds,

edge computing emerges as a new paradigm, offering connectivity, storage, and computation resources at the proximity of end users [2], such that data can be processed at edge devices, including base stations and edge routers, usually referred to as *cloudlets*.

A. The Things-Edge-Cloud Architecture

Recently, minds of infrastructure vendors, application designers and network providers have converged to a layered architecture [2] in Fig. 1: i) IoT layer, consists of “things” that harvest data, ii) edge layer, consists of cloudlets, *i.e.*, distributed servers mounted on edge routers or base stations, with moderate amounts of storage and processing power, and iii) cloud layer, consists of high-performance servers that are designed to execute computation-heavy tasks.

In this architecture, cloudlets function as both a supplement (an accessing gateway) and an alternative (the surrogate processing server) to the cloud. For light-loaded processing tasks, the application-level latency can be reduced to the mere sum of a *cloudlet access delay* (CAD) and a processing delay at the cloudlet. Compared with the latter, which depends on hardware configuration and runtime processing load, CAD is practically zero, as cloudlets are expected to provide one-hop access to most of end devices in urban settings. Consequently, existing literature on application provisioning mainly focus on minimizing the processing delay, studying resource allocation [3], application placement [2] and task offloading [4] schemes.

B. Cloudlet Access in Hostile Environments

While access to cloudlets is pervasive in the aforementioned urban settings, it can be difficult in challenging, even hostile environments. For instance, in the tactical edge [5], end devices carried by soldiers have to transmit with minimum power to avoid enemies’ detection, relying on multi-hop forwarding to delivery battlefield information to cloudlets, which are usually mounted in the command unit for tactical coordination [5], [6]. On the other hand, in post-disaster search-and-rescue missions [7], close-up pictures in residents’ mobile phones, and surveillance data in IoT devices have to be merged at the cloudlets to build post-disaster maps [8].

In both cases, Internet access, or even electrical power, may not be available, so battery-operated devices have to rely on device-to-device (D2D) communication for data transmission. Meanwhile, a limited number of cloudlets may be (or have become) the only equipment powerful enough to run life-critical applications, whose quality of service downgrades drastically as application-level latency increases.

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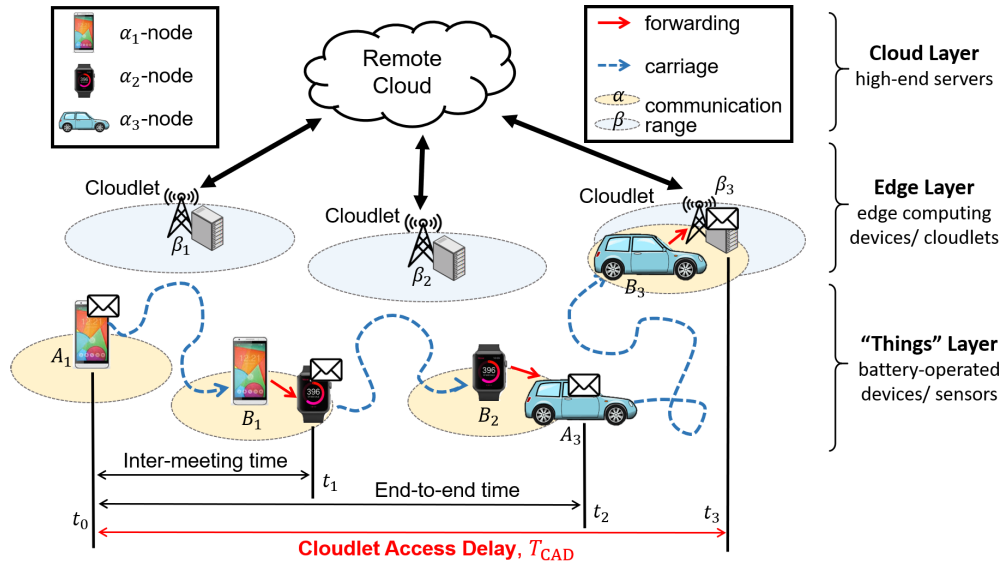


Fig. 1. Cloudlet access and packet delivery for applications in hostile environments.

Consequently, unlike typical edge computing scenarios, CAD is not negligible, but rather significant, and hence two open questions rise naturally:

- What is the expected cloudlet access delay (CAD) in hostile environment?
- How to effectively deliver data to cloudlets in such challenging scenarios?

At first glance, the cloudlet access problem bears several resemblances with those in mobile ad-hoc networks (MANETs) [9], [10] and delay-tolerant networks (DTNs) [11], [12]. Specifically, they all feature store-and-forward (SCF) actions, multi-hop communications and power-conserving objectives - in essence, an end device encapsulates harvested data into a packet, which then traverses the network through SCF actions of end devices, until (a copy of) the packet first reaches the destination. However, unique settings of the challenging military/post-disaster environment bring distinct challenges to the cloudlet access problem.

C. Unique Challenges of the Cloudlet Access Problem

From the perspective of *network settings*, due to the limitations of terrain and damaged infrastructure, end devices may use any possible communication method for data transmission, including but not limited to Bluetooth, ZigBee, LTE-D2D, etc. In other words, there is no predetermined communication or routing scheme, as opposed to carefully engineered SCF mechanisms in MANETs. Such heterogeneity demands for a *generic* modeling approach to uncover the impact factors on CAD, such that results apply to various settings.

From the perspective of *performance metrics*, CAD differs from *end-to-end delay* in MANETs and DTNs [12] by definition. As there may be multiple cloudlets in the system, CAD measures the *minimum time* it takes to reach *any cloudlet*, e.g., the CAD of packet ϵ generated at t_0 is $T_{CAD} = (t_3 - t_0)$, where t_3 is the time when a copy of ϵ reaches cloudlet β_3 , as illustrated in Fig. 1. Contrarily, end-to-end delay measures the time interval that a packet travels from the source to a

specific destination. So in a more general sense, CAD is the minimum of multiple end-to-end delays, which adds multiple layers of difficulties to the cloudlet access analysis.

From the perspective of *requirements*, applications such as search-and-rescue and tactical coordination, are usually associated with a stringent delay bound. On the contrary to the delay tolerant nature of DTNs and MANETs, data delivered after the delay bound will be useless, so it is crucial for application developers to grasp characteristics of the expected CAD *a priori*. Moreover, provisioning for delay-critical emergency applications demands for *effective* packet delivery schemes, to avoid futile transmissions in challenging environments.

D. Our Approach and Contributions

To address the cloudlet access problem with delay and energy consumption challenges, this paper focuses on the relative *movements* of a packet toward cloudlets, such that performances of the cloudlet access scheme can be analyzed and improved, to support emergency applications in challenging environments. Specifically,

- we propose a setting-independent *packet mobility model*, by which the CAD, successful access rate (SAR), and energy consumption are quantitatively studied;
- we identify sub-martingale, martingale, and super-martingale types of packet mobilities, each of which exhibits distinct CAD characteristics, scaling either linearly or quadratically, with respect to the initial distance between an end device and the nearest cloudlet;
- we design a packet shedding scheme for end devices, which drops futile data packets early at their generation. Simulation shows that it saves 24% energy and reduces 2% CAD on average, with a mere drop of 4% in SAR.

The rest of this paper is organized as follows. We first review existing literature in Section II. Then, we illustrate the *packet mobility model* in Section III. In Section IV, we show that packet mobilities in the cloudlet access scenario can be

categorized into three types, exhibiting either sub-, super-, or martingale-like properties. By this categorization, bounds on the expected CAD and SAR are analyzed in Section V. Based on these findings, we design a packet shedding scheme to reduce the average energy consumption in Section VI. Finally, the paper is concluded in Section VII. Major proofs are gathered in the Appendices for readers' references.

II. RELATED WORKS

Though the significant role of cloudlets has long been recognized [7] in hostile environments, research on this direction mainly explored the architecture of tactical edge [5], emergency applications [13], and cloudlet platforms [14]. Nonetheless, a fundamental property of the cloudlet scheme, *i.e.*, the cloudlet access delay (CAD), remain largely unknown, which hinders the design of effective packet delivery schemes for such applications.

On the avenue of delay analysis, there have been extensive studies on intermittently-connected MANET, which can be broadly categorized into one-hop and multi-hop types. The former branch evolves around the analysis of *inter-meeting time*, measuring the length of the time interval between two possible data transmissions. In our case, as illustrated in Fig. 1, the inter-meeting time equals to one-hop delay, as the transmission delay and propagation delay are usually negligible compared to the inter-meeting time of end devices. To this end, two basic models have been established, that is, the exponential inter-meeting time model [10] under random waypoint node mobility [15], which is corroborated by simulations; and the power law-tailed inter-meeting time model [16], which is obtained through real-world mobility traces. With the inter-meeting time as a building block, the latter branch studies the entire packet delivery process. On this avenue, Groenevelt et al. [10] evaluated the packet delivery delay through a Markov chain model in MANET. Under the same assumption, Zhang et al. [12] analyzed the end-to-end delay through ordinary differential equations, with respect to an independently and identically distributed (i.i.d.) inter-meeting time for networks employing epidemic routing. By a generalized model, Picu and Spyropoulos [17] analyzed the delay performance of MANET employing epidemic routing schemes.

Despite these efforts, existing results on end-to-end delay do not directly apply to the cloudlet access problem discussed here. Firstly, the exponential inter-meeting time assumption may not necessarily hold [16], which calls for a new modeling approach irrespective of inter-meeting times, to understand the characteristics of CAD in hostile environments. Secondly, CAD differs from end-to-end delay by definition, as illustrated in Fig. 1. Therefore, we propose a generic *packet mobility model* to analyze CAD, which allows us to achieve more refined results than existing analysis. Specifically, CAD can be evaluated against a simple quantity, *i.e.*, the distance between the generation location of a data packet and the nearest cloudlet. In this sense, result in [18] can be viewed as the ensemble mean in our model.

On the avenue of emergency application provisioning, which necessitates fast and energy-efficient packet delivery,

there have been study for MANETs and wireless sensor networks. Targeting at reducing end-to-end delay, packet forwarding mechanisms, including geographic routing [19], mobility-based node selection [20], [21], can provide satisfying delay performances for MANETs. However, their capability to sustain emergency applications (with stringent delay bounds) is still unknown. To this end, we define the SAR metric to quantify the likelihood of satisfying a delay bound, by which heuristic schemes such as [20] and [21] can be evaluated with respect to specific delay bounds of applications. On the other hand, energy-efficient packet delivery schemes for wireless sensor networks (see [22] and references therein) assume a connected network and are hence not suitable for the challenging environment discussed here. To address this concern, we propose a prediction-based *packet-shedding* mechanism, which discards the packets that (with a high probability) cannot be delivered on-time. Compared to existing probabilistic schemes, packet shedding with geographic-based heuristic routing [21] induces much less overhead, while achieving a better delay performance. Moreover, the proposed mechanism can be further extended to accommodate generic routing schemes.

III. PROBLEM FORMULATION

In this section, we outline the network model for hostile environments, and present the *packet mobility model* to capture the delivery process of a data packet ϵ to a cloudlet, such that the cloudlet access problem can be formulated and examined with formally defined performance metrics.

A. Network and Communication Models

Consider a rectangular geographical region $\mathcal{N} = [0, L]^2$. There are two types of nodes in the system, battery-operated mobile α -nodes, and infrastructure β -nodes running cloudlet services. Let $\mathcal{A} = \{\alpha_0, \dots, \alpha_{n-1}\}$ denote the set of n α -nodes. The cloudlet applications are installed on edge computing platforms [8], denoted here as β -nodes, which may also act as wireless gateways if their access to the Internet is intact. We assume these β -nodes $\mathcal{B} = \{\beta_0, \beta_2, \dots, \beta_{m-1}\}$ are uniformly located,¹ dividing the area of interest $\mathcal{N}$ into $l \times l$ cells, where $l \leq L$ is the side length of a cell and $m = \lfloor \frac{L}{l} \rfloor^2$ is the number of functioning β -nodes in the region $\mathcal{N}$.

Due to the limitations of terrain, post-disaster conditions, or military constraints, access to the Internet, or even to the power grid may not be available, so α -nodes in region $\mathcal{N}$ have to rely on short-range communication measures, such as WiFi, Bluetooth, or device-to-device (D2D), for data transmission. Consequently, an α -node, *e.g.*, α_0 , can only communicate with another (α - or β -)node within its transmission range r , as indicated by the shaded disk in Fig. 2a.

Considering that data harvested by α -nodes need to be processed at β -nodes (cloudlets), *e.g.*, in the afore-mentioned GigaPan [8] application, we focus on the α -to- β node direction of the data delivery process, because power supply (and hence

¹In practice, β -nodes may not be uniformly distributed, but the analysis results apply to most of the conditions, especially for large cell side-length l , *i.e.*, the remaining β -nodes are sparsely distributed over the region.

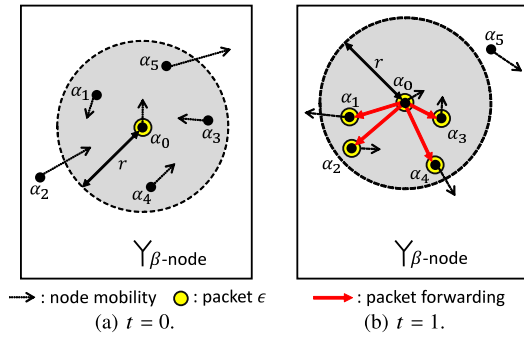


Fig. 2. An example of SCF actions.

communication range) in the reverse β -to- α direction is not a problem due to the co-location of β -nodes and infrastructure.

To closely examine the cloudlet access process, we follow the path of a single packet ϵ . Let time t proceed in discrete steps, that is, $t \in \mathcal{T} = \{0, 1, 2, \dots\}$. Without loss of generality, we set the length of each time step to 1 second. During each time step, α -nodes move and may transmit data if two nodes run into the communication range of each other. During the packet delivery process, which is a sequence of store-carry-forward (SCF) actions [12], the system (*i.e.*, rectangular region $\mathcal{N}$) is observed at the end of each time step. Let $\vec{X}_{\alpha_k}(t) \in \mathbb{R}^2$ denote the geographical location of node α_k at time t . Let $1_{\alpha_k}^\epsilon(t)$ represent the carrying status of node α_k with respect to packet ϵ at time t . For instance in Fig. 2a, $1_{\alpha_0}^\epsilon(0) = 1$, because node α_0 has packet ϵ locally stored at time $t = 0$. As we focus on the delivery of a single data packet ϵ , we omit the superscript ϵ and write $1_{\alpha_k}(t)$ instead.

Without loss of generality, let packet ϵ be generated by α_0 at time $t = 0$. Initially, $1_{\alpha_0}(0) = 1$, and $1_{\alpha_k}(0) = 0$ for all $k \neq 0$. In the next time step, ϵ is *carried* by α_0 to a new location, as shown in Fig. 2a. So at the end of time step $t = 1$, $1_{\alpha_k}(1) = 1$ for $k \in \{0, 1, \dots, 4\}$, as α_0 has forwarded ϵ to all of the nodes that are in contact with it.

B. Definition of Packet Mobility

While schemes for delivering data packets can be implemented in various ways [12], [23], how to characterize the travel distance, speed, and corresponding delivery/access time of a packet still does not have an obvious answer. Therefore, we propose a *packet mobility* model, which captures the movement of data packets with respect to the nearest β -node, such that CAD and energy consumption of a packet delivery process can be quantitatively defined base on this process.

Definition 1: Let $\vec{X}_\epsilon(t) := \{\vec{X}_{\alpha_k}(t) : 1_{\alpha_k}(t) = 1\}$ be the locations of the packet-carrying α -nodes at time $t \in \mathcal{T}$. Then, **packet mobility** is defined as the collection $\{\vec{X}_\epsilon(t)\}_{t \in \mathcal{T}}$.

Though $\{\vec{X}_\epsilon(t)\}_{t \in \mathcal{T}}$ encodes the full information of packets' movements, it is difficult to analyze directly. Hence, we focus on the most important aspect, *i.e.*, the packet *propagation speed*.

Definition 2: Let $\vec{X}_{\beta_j} \in \mathbb{R}^2$ be the location of the β -node β_j , and define the distance-to-destination of packet ϵ as

$$L_\epsilon(t) := \min_{\vec{X}_{\alpha_k}(t) \in \vec{X}_\epsilon(t), \beta_j \in \mathcal{B}} \left\| \vec{X}_{\alpha_k}(t) - \vec{X}_{\beta_j} \right\|, \quad (1)$$

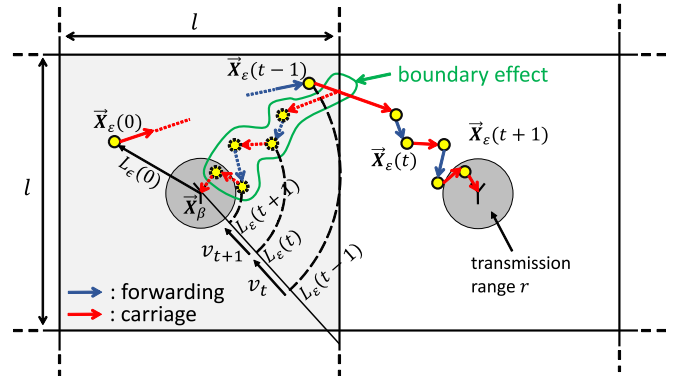


Fig. 3. The boundary effect.

where $\|\cdot\|$ is the Euclidean distance measure. The **propagation speed** of packet ϵ at time t is defined as the increment of distance-to-destination over t , *i.e.*, $v_\epsilon(t) := L_\epsilon(t) - L_\epsilon(t-1)$.

The packet propagation speed $v_\epsilon(t)$ directly shows the trend of a packet mobility, in the sense that when $v_\epsilon(t) < 0$, ϵ moves toward the destination, and moves away from the destination when $v_\epsilon(t) > 0$. Fig. 3 illustrates the relation among packet mobility $\{\vec{X}_\epsilon(t)\}_{t \in \mathcal{T}}$, distance $L_\epsilon(t)$, and propagation speed $v_\epsilon(t)$ when a single-copy routing scheme is employed.

In hostile environments, the total number (m) of functioning cloudlets, or β -nodes, is usually quite small, which means in our model, the side length of a cell l is large. Consequently, packet ϵ will most likely be delivered to the cloudlet that is closest to its generation location, *e.g.*, cloudlet β located at $\vec{X}_\beta$ in Fig. 3. In this case, Eq. (1) can be rewritten as

$$L_\epsilon(t) := \min_{\vec{X}_{\alpha_k}(t) \in \vec{X}_\epsilon(t)} \left\| \vec{X}_{\alpha_k}(t) - \vec{X}_\beta \right\|, \quad (2)$$

where $\beta = \arg \min_{\beta_j \in \mathcal{B}} \left\| \vec{X}_{\alpha_0}(0) - \vec{X}_{\beta_j} \right\|$ represents the nearest β -node to packet ϵ 's generation point.

On the other hand, when m is large, or there are multiple functioning β -nodes nearby, packet ϵ may be carried or forwarded across the cell boundary, and eventually delivered to a β -node in another cell, as depicted in Fig. 3. Due to the homogeneous setting, the trajectory of ϵ after it crosses the boundary can be analyzed as if ϵ was reflected by the boundary (see the dotted arrows in Fig. 3), and bounced back to the cell where it was originated. Therefore, it is sufficient to consider the nearest β -node (denoted as β) in the analysis, *i.e.*, Eq. (2) still holds, expect for the impacts of boundary reflection on the packet mobility (*i.e.*, $L_\epsilon(t)$ and $v_\epsilon(t)$) when packet ϵ crosses the cell boundary. We refer to this as the *boundary effect*. It can be anticipated that boundary effect will accelerate the delivery process, as we will show in Sec. V-B, so the single β -node case $m = 1$ depicted in Eq. (2) marks the worst-case scenario of cloudlet access.

C. Problem Formulation

To provision for applications in challenging environments, it is necessary to analyze performances of cloudlet access, with respect to delay and energy consumption, respectively.

From the perspective of application delay, the time to deliver packet ϵ to a β -node, that is, the cloudlet access delay T_{CAD}^ϵ is rigorously defined as follows.

Definition 3: Cloudlet access delay (CAD) of packet ϵ is defined as the time duration from the instant ϵ is generated (denoted as t_ϵ) until it reaches at least one β -node, that is,

$$T_{CAD}^\epsilon := \inf_{t \in \mathcal{T}} \{t : L_\epsilon(t) < r\} - t_\epsilon. \quad (3)$$

Unlike the end-to-end delay [12], T_{CAD}^ϵ measures the delay on the mobile-to-cloudlet direction (see Fig. 1), while the inverse direction is not the concern of this paper, as we focus on the provisioning for emergency applications, such as GigaPan. To support such applications with a limited number (m) of cloudlets, it is important to examine whether a certain *delay bound*, τ , can be satisfied. For example, first responders need to reach an emergency location within 5 minutes to save a victim's life after he/she requested medical assistance [24], so the delay bound for the cloudlet access $\tau = 300$ time slots. In other words, if $T_{CAD}^\epsilon > \tau$, this attempt to deliver packet ϵ is futile, as the ϵ will not reach any cloudlet in time.

In addition to strict delay requirements, end devices, *i.e.*, battery-operated α -nodes are usually subjected to strict energy constraints, since their access to stable electricity supply is highly restricted. Therefore, we define a *power draw* metric to quantify the amount of transmission power spent on the packet delivery process to the cloudlets.

Let $P_\epsilon(t, j)$ denote the transmit power consumed when the j^{th} copy of ϵ is forwarded at time $t \in \mathcal{T}$ from a packet-carrying α -node α_{k_1} to (i) another α -node α_{k_2} or (ii) the β -node. Note that $P_\epsilon(t, j) = 0$ if ϵ is not forwarded, but carried during time slot t . We assume the packet forwarding actions stop at time T_{CAD}^ϵ , when ϵ is delivered to the β -node.²

Definition 4: The per-delivery power draw is defined as the summation of transmission power consumed to deliver packet ϵ , that is,

$$P_\epsilon = \sum_{t=t_\epsilon+1}^{T_{CAD}^\epsilon} \sum_{j=1}^{|\vec{\mathbf{X}}_\epsilon(t)|} P_\epsilon(t, j). \quad (4)$$

With T_{CAD}^ϵ and P_ϵ at hand, we restate our research question: Consider a set $\mathcal{A}$ of α -nodes in the region of interest $\mathcal{N}$, and a β -node located at $\vec{\mathbf{X}}_\beta \in \mathcal{N}$. Let packet ϵ be generated by node α_0 at time $t = 0$, associated with delay constraint τ . For packet mobility $\{\vec{\mathbf{X}}_\epsilon(t)\}_{t \in \mathcal{T}}$,

- i) What is the expected cloudlet access delay $\mathbb{E}(T_{CAD}^\epsilon)$?
- ii) How to minimize power-draw P_ϵ , without downgrading the delivery rate $\mathbb{P}(T_{CAD}^\epsilon < \tau)$?

IV. CHARACTERIZATION OF PACKET MOBILITIES

By and large, packet mobility $\vec{\mathbf{X}}_\epsilon(t)$ is a comprehensive model, because it contains full information to derive both

²Considering that cloudlets are mounted on the infrastructure, and $\mathcal{B}$ refers to the functioning β -nodes in the hostile environment, it is reasonable to assume that β -nodes do not need to conserve energy, and hence can broadcast a message to let all packet-carrying α -nodes at T_{CAD}^ϵ to drop ϵ immediately, *i.e.*, $|\vec{\mathbf{X}}_\epsilon(T_{CAD}^\epsilon + 1)| = 0$. Moreover, as the queuing and processing delay (usually several ms) are not comparable to the inter-meeting time of nodes (seconds to minutes), they are not considered in the CAD analysis.

long-term and transient properties of the packet delivery process. However, it is difficult to track the evolution of the process for all possible SCF combinations of node mobilities and packet forwarding schemes. To tackle this challenge, we examine the building block of packet mobility, that is, the distance-to-destination $L_\epsilon(t)$, such that various packet mobility processes can be characterized with properties of $L_\epsilon(t)$. In particular, we find that, packet mobility under frequently seen SCF combinations exhibits either super-martingale, martingale, or sub-martingale properties in their corresponding $\{L_\epsilon(t)\}_{t \in \mathcal{T}}$ processes, each leading to a distinct CAD behavior, which will be discussed in the next section.

A. SCF Combinations in the Cloudlet Access Scenario

Movements of a packet are perpetuated by two types of SCF actions: node movement, which can be characterized by mobility models [15] or depicted by traces such as [25]; and packet forwarding, which is governed by forwarding/routing schemes, such as randomized routing [23] and epidemic routing [12]. We consider the possible SCF combinations of four mobility settings (T1–T4) and three forwarding schemes (R1–R3) for the cloudlet access problem.

- [T1] **Random walk (RW)** [15]: At every time slot $t \in \mathcal{T}$, each α -node selects to move to a new location by randomly choosing a speed and a direction from uniform distributions with ranges $[v_{min}, v_{max}] = [1, 10]$ m/s and $[0, 2\pi]$ *rads*, respectively.
- [T2] **Random waypoint (RWP)** [15]: Every α -node pauses for a time chosen randomly from range $[3, 8]$ s, picks a random destination, and then moves there with a random speed within $[v_{min}, v_{max}] = [1, 10]$ m/s. These steps are repeated until the network lifetime ends.
- [T3] **ncsu/statefair** [25]: GPS traces of 19 visitors during two North Carolina state fair events on Aug. 26, 2006 and Apr. 18, 2008.
- [T4] **ncsu/disneyworld** [25]: GPS traces of 41 volunteers that visited the Disney World in Florida during holiday seasons of 2006 and 2007.

Among these, T1 and T2 are the simplest and most frequently used synthetic node mobility models in mobile network analysis, while T3 and T4 are real-world traces in public venues, mimicking trajectories of first responders and soldiers in hostile environments.

As for packet forwarding, we consider schemes in the simplest form, due to the limitation of challenging environments. For instance, in post-disaster scenarios, devices have to fall-back to basic forwarding schemes available to devices of any make, type, and service providers. In this case, complex routing schemes such as RPL [26], OLSR [27], and AODV [27], which either requires end-to-end connectivity or the storage of lengthy routing tables, are not discussed.

- [R1] **Direct contact** [9], [28]: The packet-generating node α_0 carries ϵ , and delivers ϵ directly to the β -node when it comes into the transmission range of β . In fact, there is no forwarding action in this scheme, which is adopted as the baseline case to derive other results.

- [R2] **Randomized routing** [23]: The packet-carrying α -node forwards ϵ to one of encountered neighbors selected uniformly at random, until ϵ reaches the β -node.
- [R3] **Epidemic routing** [12]: The packet-carrying α -node replicates ϵ to *all* the encountered neighbor nodes until ϵ reaches the β -node. This scheme achieves the *best* possible delay performance, but also consuming the most transmission energy.

Altogether, there are 12 T^*+R^* combinations in the cloudlet access scenarios. In the next subsection, we provide the mathematical tool to examine their individual properties.

B. Building Block $\{v_\epsilon(t)\}_{t \in \mathcal{T}}$ of a Packet Mobility

At time $t = 0$, packet ϵ is generated by an α -node α_0 , whose current distance from the cloudlet β -node is $L_\epsilon(0)$. Then at time $t \in \mathcal{T}$, packet ϵ 's distance to destination can be written as $L_\epsilon(t) = L_\epsilon(0) + \sum_{k=1}^t v_\epsilon(k)$, where $L_\epsilon(0)$ is determined at ϵ 's generation, and each $v_\epsilon(k)$ is a random variable, jointly determined by the movements and forwarding actions of α -nodes. Then, characteristics of the packet mobility, or equivalently random process $\{L_\epsilon(t)\}_{t \in \mathcal{T}}$, are mainly determined by random variable $v_\epsilon(t)$, who has the following properties.

- Random variable $v_\epsilon(t)$ is bounded, and it takes value from range $[-(v_{node} + r), (v_{node} + r)]$, where v_{node} is the maximum speed of a mobile node.
- Probability space $(\Omega, \mathcal{F}, \mathbb{P})$ of $v_\epsilon(t)$ is determined by the T^*+R^* combination.

Without loss of generality, let $v_\epsilon(0) = 0$. Then we have the following lemma on $\{L_\epsilon(t)\}_{t \in \mathcal{T}}$.

Lemma 1: Given $\mathbb{E}(v_\epsilon(t)|\mathcal{F}_{t-1}) = 0$ (< 0 , and > 0 , respectively) for all $t \in \mathcal{T}$, process $\{L_\epsilon(t)\}_{t \in \mathcal{T}}$ is a martingale (sub-martingale, and super-martingale, respectively) with respect to the propagation speed $\{v_\epsilon(t)\}_{t \in \mathcal{T}}$.

Proof of Lemma 1 can be found in Appendix A. It provides a criterion on the expected speed $\mathbb{E}(v_\epsilon(t)|\mathcal{F}_{t-1})$, identifying three patterns on process $\{L_\epsilon(t)\}_{t \in \mathcal{T}}$, each of which corresponds to a type of packet mobility, defined as follows.

Definition 5: If process $\{L_\epsilon(t)\}_{t \in \mathcal{T}}$ is a martingale (sub- and super-martingale, respectively), we say $\bar{X}_\epsilon(t)$ is a **martingale (sub- and super-martingale)-type packet mobility**.

Then we have the following corollaries for the T1+R1 combination, and Infostation model [9] (special case of T1)+R2 combination, the latter of which illustrates a typical cloudlet access scenario for emergency vehicles. Proof of Corollary 1 and 2 can be found in Appendix B.

Corollary 1: When α -nodes in $\mathcal{A}$ move according to RW mobility (T1) in region $\mathcal{N}$, and employ direct contact (R1) forwarding, the packet mobility is of sub-martingale type.

Corollary 2: Consider a set $\mathcal{A}$ of α -nodes distributed uniformly along a line segment, by which β -nodes are evenly spaced. When α -nodes move according to RW mobility (2-D version of T1), and randomized forwarding (R2) is employed, the resulting packet mobility is of martingale type.

Note that Definition 5 is not a full partition, which means there exist packet mobilities that do not fall into any of the

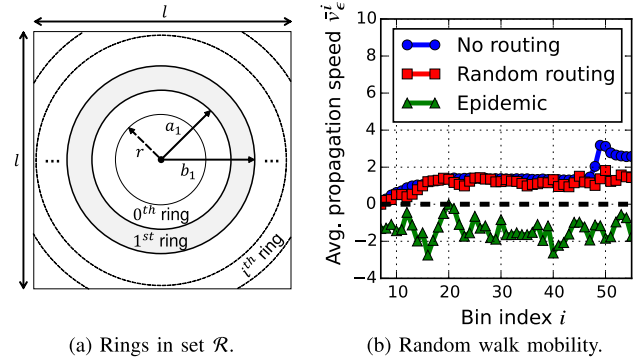


Fig. 4. Evaluation of the propagation speed $\bar{v}_\epsilon^i$.

three categories. Nonetheless, as will be discussed in the next subsection, general SCF combinations in this paper can be examined against a relaxed criterion.

C. Characterization of Packet Mobilities

The first challenge of directly using Definition 5 lies in examining $\mathbb{E}(v_\epsilon(t)|\mathcal{F}_{t-1})$ against zero, who is a bounded random variable. By definition, r. v. $\mathbb{E}(v_\epsilon(t)|\mathcal{F}_{t-1})$ takes value $\mathbb{E}(v_\epsilon(t)|L_\epsilon(t-1) = x)$, when the continuous random variable $L_\epsilon(t-1)$ takes value $x \in (r, l]$.

Since interval $(r, l]$ is continuous, we consider the discrete case as an approximate. Specifically, we divide the $l \times l$ cell area into rings $\mathcal{R} = \{(a_0, b_0], (a_1, b_1], \dots\}$, where $a_i \geq r$ and $b_i > a_i$ are the inner and outer diameters of the i -th ring (see Fig. 4a), which serve as the “bins” for $L_\epsilon(t-1)$. Then we have the following characterization of packet mobilities, with a relaxed criterion.

Definition 6: A packet mobility is **sub-martingale-like, martingale-like, or super-martingale-like** if for any $(a_i, b_i] \in \mathcal{R}$, the expected packet propagation speed $\mathbb{E}(v_\epsilon(t)|L_\epsilon(t-1) \in (a_i, b_i])$ in the i -th ring is greater than, equal to, or smaller than zero, respectively.

Definition 6 provides a viable way to examine packet mobilities, and when the partition of rings is more refined, i.e., large $|\mathcal{R}|$, the stricter is Definition 6, and closer it is to Definition 5. However, when packet forwarding actions are taken into account, e.g., R2 and R3, the expected propagation speed also depends on the locations of non-carrying α -nodes at time $t-1$, hindering the direct application of Definition 6. To tackle this challenge, we leverage a numerical method to estimate $\mathbb{E}(v_\epsilon(t)|L_\epsilon(t-1) \in (a_i, b_i])$ specified as follows.

The main idea of this numerical method is to estimate $\mathbb{E}(v_\epsilon(t)|L_\epsilon(t-1) \in (a_i, b_i])$ with the ensemble mean. Specifically, we generate a large number of packet mobility realizations $S \subset \Omega$, each sample of which is generated with a random initial value $\bar{X}_\epsilon(0)$. For every time slot $t \in \mathcal{T}$, random variable $\mathbb{E}(v_\epsilon(t)|L_\epsilon(t-1) \in (a_i, b_i])$ is a mapping from the sample space Ω to $\mathbb{R}^+$. For the RW mobility model T1, and the RWP model T2, the moving speed of α -nodes is drawn uniformly at random from the same distribution. For the cases of mobility traces T3 and T4, though α -nodes move according to people's free will, analysis in [25] showed that node/human mobility exhibits salient characteristics of Lévy flights, whose

flight distance per unit time (or equivalently the moving speed) follows from the *same* heavy-tailed distribution. On the other hand, packet forwarding scheme R1 does not contain any forwarding actions, while recipients in R2 and R3 are chosen uniformly at random. Consequently, the packet propagation speed $v_\epsilon(t)$ in all the 12 SCF combinations is independent, and identically distributed (i.i.d.) over the sample space Ω . Moreover, by the Markov property of RW (T1), RWP (T2), and Lévy walk (T3 and T4), the exact value of time t does not matter, as long as the current packet mobility state $\vec{X}_\epsilon(t-1)$ is known. Then by the law of large numbers, the expected packet propagation speed $\mathbb{E}(v_\epsilon(t)|L_\epsilon(t-1) \in (a_i, b_i])$ can be estimated with the ensemble mean

$$\bar{v}_\epsilon^i := \frac{\sum_{\omega \in S, t \in T, L_\epsilon(t-1)(\omega) \in (a_i, b_i]} v_\epsilon(t)(\omega)}{\sum_{\omega \in S, t \in T} \mathbf{1}_{L_\epsilon(t-1)(\omega) \in (a_i, b_i]}}, \quad (5)$$

where $\mathbf{1}_{(\cdot)} \in \{0, 1\}$ is the indicator function. Moreover, $\lim_{|S| \rightarrow \infty} \bar{v}_\epsilon^i = \mathbb{E}(v_\epsilon(t)|L_\epsilon(t-1) \in (a_i, b_i])$. We refer to $\bar{v}_\epsilon^i$ as the average propagation speed for the i -th bin. According to Eq. (5), we collect numerical samples $v_\epsilon(t)(\omega)$ in OmNet++'s Inetmanet-2.0 to obtain $\bar{v}_\epsilon^i$ for the 12 SCF combinations, with configuration detailed in Table I.

Under this setting, we first examine the packet mobility induced by the random walk mobility (T1) with direct contact (R1, blue line with round markers), random routing (R2, red line with square markers), and epidemic routing (R3, green line with triangle markers), respectively. The corresponding average speeds $\{\bar{v}_\epsilon^i\}_{0 \leq i \leq |\mathcal{R}|}$ for the i -th bin are shown in Fig. 4b. A key observation is that the average speeds among all the bins are either all positive or all negative, indicating a consistent packet mobility type among the realizations. For instance, the baseline case T1+R1 induces a sub-martingale-like packet mobility ($\bar{v}_\epsilon^i > 0$ for all i), which complies with Corollary 2. For the case of T1+R2, all of $\{\bar{v}_\epsilon^i\}_{0 \leq i \leq |\mathcal{R}|}$ are positive, so r.v. $\mathbb{E}(v_\epsilon(t)|L_\epsilon(t-1) \in (a_i, b_i]) > 0$ for all $i \in [1, |\mathcal{R}|]$, and consequently T1+R2 induces a sub-martingale-like packet mobility. On the contrary, T1+R3 leads to a super-martingale-like packet mobility, as $\bar{v}_\epsilon < 0$.

Observing the consistent sign among the mean propagation speed of different bins, it is reasonable to evaluate the arithmetic mean of all the bins instead, which we refer to as the *overall average speed*,³ that is, $\bar{v}_\epsilon := \frac{1}{|\mathcal{R}|} \sum_{i=0}^{|\mathcal{R}|-1} \bar{v}_\epsilon^i$.

The overall average speed and the corresponding packet mobility type of the 12 SCF combinations are presented in Table II. By Definition 6, the packet mobilities for the cloudlet access in challenging environments are either sub-, super-, and martingale-like types. For these types, process $\{L_\epsilon(t)\}_{t \in T}$ will have nice stochastic properties, which allows us to derive performances of cloudlet access, specifically, the cloudlet access delay T_{CAD}^ϵ and per-delivery power draw P_ϵ .

³Technically, the expected propagation speed $\mathbb{E}(v_\epsilon(t)) = \sum_{0 \leq i \leq |\mathcal{R}|} \mathbb{E}(v_\epsilon(t)|L_\epsilon(t-1) \in (a_i, b_i]) \mathbb{P}(L_\epsilon(t-1) \in (a_i, b_i])$, so the mean needs to be estimated with $\sum_{0 \leq i \leq |\mathcal{R}|} \bar{v}_\epsilon^i \mathbb{P}(L_\epsilon(t-1) \in (a_i, b_i])$. However, as $\gamma = b_i - a_i \ll \infty$ and $L_\epsilon(0)$ is generated uniformly at random, we assume $\mathbb{P}(L_\epsilon(t-1) \in (a_i, b_i]) = \frac{1}{|\mathcal{R}|}$, and employ $\bar{v}_\epsilon$ as the ensemble mean of $\{v_\epsilon(t)\}$.

V. ANALYSIS OF THE CLOUDLET ACCESS DELAY (CAD)

In challenging environments, CAD is the most influential performance metric, because data for emergency applications are required to be delivered within a stringent delay bound τ . In this section, we discuss the expected CAD $\mathbb{E}(T_{CAD}^\epsilon)$ with respect to the three types of packet mobilities, and then provide bounds on SAR $R_{L_0}^\tau$ with respect to the packet propagation speed v_ϵ and initial distance-to-destination $L_\epsilon(0)$. Specifically, we start with the special case of $r \ll l \simeq L$, where there is only $m = 1$ β -node in region $\mathcal{N}$, and then expands to the general case of multiple ($m > 1$) cloudlets, where the boundary effect is discussed. We find that each type of packet mobility is associated with a distinct bound of $\mathbb{E}(T_{CAD}^\epsilon)$ with respect to $L_\epsilon(0)$, while the boundary effect is significant on martingale and super-martingale-like types.

A. Cloudlet Access Delay Under Unbounded Cell Coverage

Military or post-disaster environments can be extremely harsh to cloudlet access, e.g., when military operations are carried out far beyond the enemy line, deploying cloudlets are costly and may even alert the enemies. Consequently the total number of cloudlets $m = |\mathcal{B}|$ is strictly limited. Therefore, we first consider the extreme case where there is only a single β -node ($m = 1$), such that the instantaneous distance-to-destination $L_\epsilon(t)$ may not be bounded.

1) *Super-Martingale-Like Packet Mobility*: Since packet ϵ has a long-term tendency to move towards the β -node, the expected delay $\mathbb{E}(T_{CAD})$ should be smaller than the other packet mobility cases. Leveraging the super-martingale-like property of process $\{L_\epsilon(t)\}_{t \in T}$, we derive bounds on $\mathbb{E}(T_{CAD})$, which complies with the intuition, and also outlines its *linear* scaling with respect to the initial condition $L_\epsilon(0)$.

Theorem 1: (Super-martingale-like: Linear Scaling) For super-martingale-like packet mobilities with independently distributed speeds $\{v_\epsilon(t)\}_{t \in T}$, the expected CAD is bounded by

$$\frac{L_\epsilon(0) - r}{-\bar{v}_\epsilon} \leq \mathbb{E}(T_{CAD}^\epsilon) \leq \frac{L_\epsilon(0)}{-\bar{v}_\epsilon}, \quad (6)$$

where $\bar{v}_\epsilon (< 0)$ is the overall average speed, and r is the transmission range of α - and β -nodes.

Proof of Theorem 1 can be found in Appendix C. The main idea is to construct a martingale sequence $\{L'_\epsilon(t)\}$ out of super-martingale $\{L_\epsilon(t)\}_{t \in T}$ with *Doob's decomposition* [29, Thm. 25.3], such that the *Optional Stopping Theorem* (OST) [29, Thm. 24.6] can be used to estimate $L'_\epsilon(T_{CAD}^\epsilon)$ at the delivery of the packet. Then the bounds can be obtained by Wald's first inequality.

The linear form $\mathbb{E}(T_{CAD}^\epsilon) = \Theta(L_\epsilon(0))$ in Theorem 1 reminds us of [18], which states that the time of (a packet) hitting the target centred in a disk area of radius a is $\Theta(a)$ under a location-based forwarding scheme. From the perspective of application, Theorem 1 extends [18] in the sense that it reveals that a multi-copy forwarding scheme, e.g., epidemic routing (R3), also leads to a delay that scales *linearly* over the initial distance. Moreover, Theorem 1 is more generic, in the sense that it provides a more relaxed sufficient condition for this

TABLE I
CONFIGURATION OF THE NUMERICAL METHODS

Scenario	Parameter	Value	Scenario	Parameter	Value
Common (All)	Time span $ \mathcal{T} $	8.64×10^5 s	Common (All)	No. of realizations $ S $	4×10^4
Common (All)	Transmission range r	25 m	Common (All)	Ring width $\gamma = b_i - a_i$	5 m
Common (All)	Cell side-length l	500 m	Synthetic (T1,T2)	No. of α -nodes per cell	100
Realistic (T3)	Region side-length L	1,000 m	Realistic (T4)	Region side-length L	18,000 m

TABLE II
PACKET MOBILITY TYPE (AND $\bar{v}_\epsilon$) OF THE 12 SCF COMBINATIONS

Node Mobility \ Routing	Direct contact (R1)	Randomized (R2)	Epidemic (R3)
Random Walk (T1)	Sub-martingale (0.10266)	Sub-martingale (0.08675)	Super-martingale (-0.04357)
Random Waypoint (T2)	Sub-martingale (0.09345)	Sub-martingale (0.08892)	Super-martingale (-0.04599)
ncsu/statefair (T3)	Sub-martingale (0.43387)	Sub-martingale (0.41572)	Super-martingale (-0.46749)
ncsu/disneyworld (T4)	Sub-martingale (2.00356)	Sub-martingale (1.97402)	Super-martingale (-0.40107)

type (linear) of scaling behavior, *i.e.*, *super-martingale-like* packet mobilities, which is not restricted to a particular routing scheme, but rather characterizes a family of SCF combinations.

2) *Sub-Martingale-Like Packet Mobility*: In this case ($\bar{v}_\epsilon > 0$), packet ϵ has the long-term tendency to move away from the cloudlet, inducing a larger $\mathbb{E}(T_{CAD}^\epsilon)$.

Theorem 2: (Sub-martingale-like: Quadratic Scaling) For identically distributed $\{v_\epsilon(t)\}_{t \in \mathcal{T}}$, the expected CAD of a sub-martingale-like packet mobility satisfies

$$\frac{(L_\epsilon(0) - r)^2}{\bar{\sigma}^2 + \bar{v}_\epsilon^2 + 2(v_{max} + r)} \leq \mathbb{E}(T_{CAD}^\epsilon) \leq \frac{(L_\epsilon(0))^2}{\bar{\sigma}^2 + \bar{v}_\epsilon^2}, \quad (7)$$

where v_{max} is the maximum speed of the α -nodes, such that $|v_\epsilon(t)| \leq v_{max} + r$, $\bar{v}_\epsilon$ is overall average speed, and variance $\bar{\sigma}^2 := \mathbb{E}(\mathbb{E}(v_\epsilon(t)^2 | \mathcal{F}_{t-1}) - \mathbb{E}^2(v_\epsilon(t) | \mathcal{F}_{t-1}))$.

Proof of Theorem 2 can be found in Appendix D. To apply the OST, we tackle the possibly infinite T_{CAD}^ϵ with an algebraic manipulation to derive bounds on $\mathbb{E}(T_{CAD}^\epsilon)$.

A clear observation from Theorem 2 is that $\mathbb{E}(T_{CAD}^\epsilon) = \Theta(L_\epsilon^2(0))$, which is *quadratic* rather than linear, indicating a much slower cloudlet access compared to the super-martingale-like case. In this sense, the $\Theta(\frac{a^2}{b})$ hitting time result in [18], where a denotes the radius of the disk region, and b denotes the transmission range in a random forwarding scenario, is a special case (ensemble mean) of Theorem 2, provided that $L_\epsilon(0)$ is chosen uniformly at random from range $(r, l]$. In other words, Theorem 2 gives the root-cause of the quadratic-form scaling law, *i.e.*, the sub-martingale-like packet mobility, and can be applied to cases of any valid $L_\epsilon(0)$ value.

3) *Martingale-Like Packet Mobility*: For this borderline case, the packet of interest tends to stay at the same distance to the cloudlet, resulting in the following bounds.

Theorem 3: (Martingale-like: Quadratic Scaling) For martingale-like packet mobilities with initial condition $L_\epsilon(0)$, the expected CAD satisfies

$$\frac{(L_\epsilon(0) - r)^2}{\bar{\sigma}^2} \leq \mathbb{E}(T_{CAD}^\epsilon) \leq \frac{(L_\epsilon(0))^2}{\bar{\sigma}^2}, \quad (8)$$

where r is the transmission range of α - and β -nodes, and $\bar{\sigma}^2$ is the variance as in Theorem 2.

Proof of Theorem 3 can be found in Appendix E. It is rather interesting to see that, the scaling behavior of martingale-like

packet mobilities is the same as the sub-martingale-like type, indicating that a strictly negative $\bar{v}_\epsilon$ is a necessary condition for a linearly scaling delay.

Putting these analysis results together with Table II, we can see that $\bar{v}_\epsilon < 0$ takes place only when the multi-copy forwarding scheme R3 is employed, despite types of node mobility. This effect is especially evident when real-world mobility traces are considered, as indicated by the relatively large speed gap between the single-copy packet forwarding schemes (R1 and R2) and the multi-copy scheme (R3). A direct implication for application design is that this non-linearity of delay over the distance should be taken into consideration when estimating the cloudlet system performances, and more aggressive forwarding schemes (*i.e.*, consuming more resource) may be required to achieve a unified quality of service. In a broader sense, the distinct scaling laws reiterated the significance of characterizing the packet mobilities with packet propagation speeds $v_\epsilon(t)$, which outlines a more applicable criterion to analyze the cloudlet access performances.

B. Effect of A Finite Cell Boundary on CAD

When the cell boundary is finite ($l \ll L$), with multiple functioning β -nodes in region $\mathcal{N}$, the expected CAD is intuitively lower than the single β -node case with unbounded cell boundaries, but will the *boundary effect* illustrated in Fig. 4a also affect the expected CAD? Considering that boundary effect has been proved to change the distribution of inter-meeting time from power-law [16] to exponential, we aim to find whether the impact is *significant*, that is, whether the scaling orders of $\mathbb{E}(T_{CAD}^\epsilon)$ in Theorem 1, 2 and 3 change due to the boundary effect.

To this end, we perform numerical simulations in OmNeT++, with a simulation update interval of 1s. The network is divided into square cells of $l \times l = (10^3 m)^2$, in which there are initially $n = 100$ α -nodes per cell. The carrier frequency is set to 2.4GHz, while the medium access control (MAC)-layer bit rate is set to 2Mbps. The transmission power is 2mW, and the signal attenuation threshold is -66dBm, while the fading model is a path-loss model with a coefficient of 2, which corresponds to the node transmission range $r = 25m$.

1) *Super-Martingale-Like Packet Mobility*: We consider the RWP node mobility and epidemic routing (T2+R3) that has $\bar{v}_\epsilon \approx -4.342 \times 10^{-2} m/s$, which falls into the super-martingale category as shown in Table II. In this case, packet ϵ tends to move towards the β -nodes over time. So it is likely that ϵ will arrive at the β -node before hitting the cell boundary, as a result of which the effect of finite boundary should be negligible. This hypothesis is validated by our numerical results, shown as the dotted green lines with triangle markers in Fig. 5a and 5b, while the lower and upper bounds are depicted by the solid blue lines with square markers and red lines with round markers, respectively. In both the $l = 1000m$ and $l = 500m$ case, Eq. (6) still holds, as well as the linear scaling order (magenta line), indicating an insignificant boundary effect.

2) *Sub-Martingale-Like Packet Mobility*: In this case, packets tend to move away from the cloudlet, towards the finite cell boundary. As discussed in Section III-B, a packet ϵ that moves across the cell border will be mirrored back into the original cell, mimicking the movements of ϵ toward the β -node in the adjacent cell. Due to the existence of multiple ($m > 1$) β -nodes, the expected CAD should be much lower than that of the single β -node case, especially when the initial distance-to-destination $L_\epsilon(0)$ is large, i.e., $L_\epsilon(0) \simeq l$. We consider a random walk mobility with a randomized routing (T1+R2). This packet mobility case is selected since it exhibits a relatively constant expected overall average speed $\bar{v}_\epsilon$ and variance $\bar{\sigma}_\epsilon^2$, which by Eq. (7) should give a clear indication on how $\mathbb{E}(T_{CAD}^\epsilon)$ grows with respect to $L_\epsilon(0)$.

From Fig. 5c and 5d, observe that the $\mathbb{E}(T_{CAD}^\epsilon)$ agrees with its lower and upper bounds in Theorem 2, and exhibits the same quadratic scaling law in Fig. 5c, when the cell-side length $l = 1000m$. When l is further reduced, however, the expected CAD becomes smaller and the quadratic scaling cannot be observed for large $L_\epsilon(0)$ (see the diverging tails in Fig. 5d). Such an observation agrees with our conjecture, indicating that the boundary effect is insignificant when $L_\epsilon(0) \ll l$, but it becomes significant when $L_\epsilon(0)$ is comparable to l .

3) *Martingale-Like Packet Mobility*: We consider the T1+R2 case depicted in Corollary 2, that is, every α -node moves at a speed v_{node} chosen uniformly at random from range $[1, 9]m/s$ [9]. From the expected CAD (green dotted line with triangle markers) in Fig. 5e, we observe that, despite the relatively large cell side-length $l = 10^4m$, which is much larger than that in Fig. 5a through 5d, the quadratic scaling behavior in Theorem 3 can only be observed up to the initial distance of $L_\epsilon(0) = 90m$. So we conclude that the boundary effect is generally significant in the martingale-like packet mobilities, even for relatively small initial distances ($L_\epsilon(0) \ll l$). Compared with the other two types, the higher sensitiveness to boundary effect is reasonable for this type, as it is a boarder-line case (with a rather strict criterion $\bar{v}_\epsilon = 0$) between the super- and sub-martingale like packet mobilities.

C. Successful Access Rate $R_{L_0}^\tau$

In hostile environments, packet delivery that exceeds the delay constraint, denoted as τ , will be futile and hence discarded by the β -node upon delivery. Therefore, application

designers need to know the delay performance of the cloudlet scheme to evaluate the QoS. For this purpose, we formally define the *successful access rate* (SAR) $R_{L_0}^\tau$ as the probability that a packet generated at the initial distance L_0 can be delivered to a β -node within the delay bound τ , that is,

$$R_{L_0}^\tau := \mathbb{P}(T_{CAD}^\epsilon < \tau | L_\epsilon(0) = L_0). \quad (9)$$

In practice, $R_{L_0}^\tau$ can be calculated as the ratio of successfully delivered packets over the total number of data packets generated from the distance L_0 . Though metrics of a similar nature have been analyzed for specific forwarding schemes, such as greedy location-based routing in DTN [30], it is unknown for general cloudlet access settings. So we first provide the following bounds outlining impacts of the initial condition and packet mobility characteristics on the SAR.

Theorem 4: (Bounds on SAR) If propagation speeds $\{v_\epsilon(t)\}_{t \in \mathcal{T}}$ of packet ϵ are i.i.d. with overall average $\bar{v}_\epsilon$ and variance $\bar{\sigma}_\epsilon^2$, then the SAR, that is, the probability that a packet can be successfully delivered within delay bound τ , satisfies

$$\begin{aligned} 1 - \exp \left\{ -\frac{([\lambda_{L_0}^\tau]^+)^2}{2(\xi_\tau + \frac{\tau + v_{max}}{3}[\lambda_{L_0}^\tau]^+)} \right\} &\leq R_{L_0}^\tau \\ &\leq \exp \left\{ -\frac{([-\lambda_{L_0}^\tau]^+)^2}{2(\xi_\tau + \frac{\tau + v_{max}}{3}[-\lambda_{L_0}^\tau]^+)} \right\}, \end{aligned} \quad (10)$$

where quantity $\lambda_{L_0}^\tau = r - L_0 - \tau \bar{v}_\epsilon$, and $\xi_\tau = \tau(\bar{\sigma}_\epsilon^2 + (\bar{v}_\epsilon)^2)$.

Eq. (10) is quite straight-forward to obtain with concentration inequalities in [31], so proof of Theorem 4 is omitted. Note that Theorem 4 does not assume any specific type, and is hence a generic result that applies to all super-, sub-, and martingale-like packet mobilities. Moreover, it is valid for every packet as long as the initial distance and the type of packet mobility are fixed, so it can be used directly (as a QoS measure) by specific applications, and the overall SAR R_{sus}^τ can be calculated by averaging over different L_0 values.

To validate these bounds, we consider RWP with epidemic routing (T2+R3) and RWP with direct contact (T2+R1) for the super- and sub-martingale-like cases, respectively. The cell side-length, number of α -nodes per-cell, and transmission range are set to $l = 1500m$, $n = 50$, and $r = 25m$, while other simulation parameters are the same as in Table I.

In all plots in Fig. 6, the SAR (solid green lines) are well-bounded by the upper and lower bounds (red and blue dashed lines, respectively) in Eq. (10). We highlight three observations: i) Bounds in Theorem 4 are tighter when the initial distance is larger, e.g., $L_\epsilon(0) = 90m$, because CAD can be as low as zero with a strictly positive probability, when packet ϵ originated too close ($r \rightarrow L_\epsilon(0)$) to the β -node. ii) There exists a critical value τ_c in super-martingale-like packet mobilities, around which the SAR changes abruptly (see the staircase shapes in Fig. 6a), due to the relatively large differences in speeds of RWP mobility. iii) Empirical SAR collected from simulation is close to its upper bound.

VI. ENERGY-EFFICIENT DATA DELIVERY IN

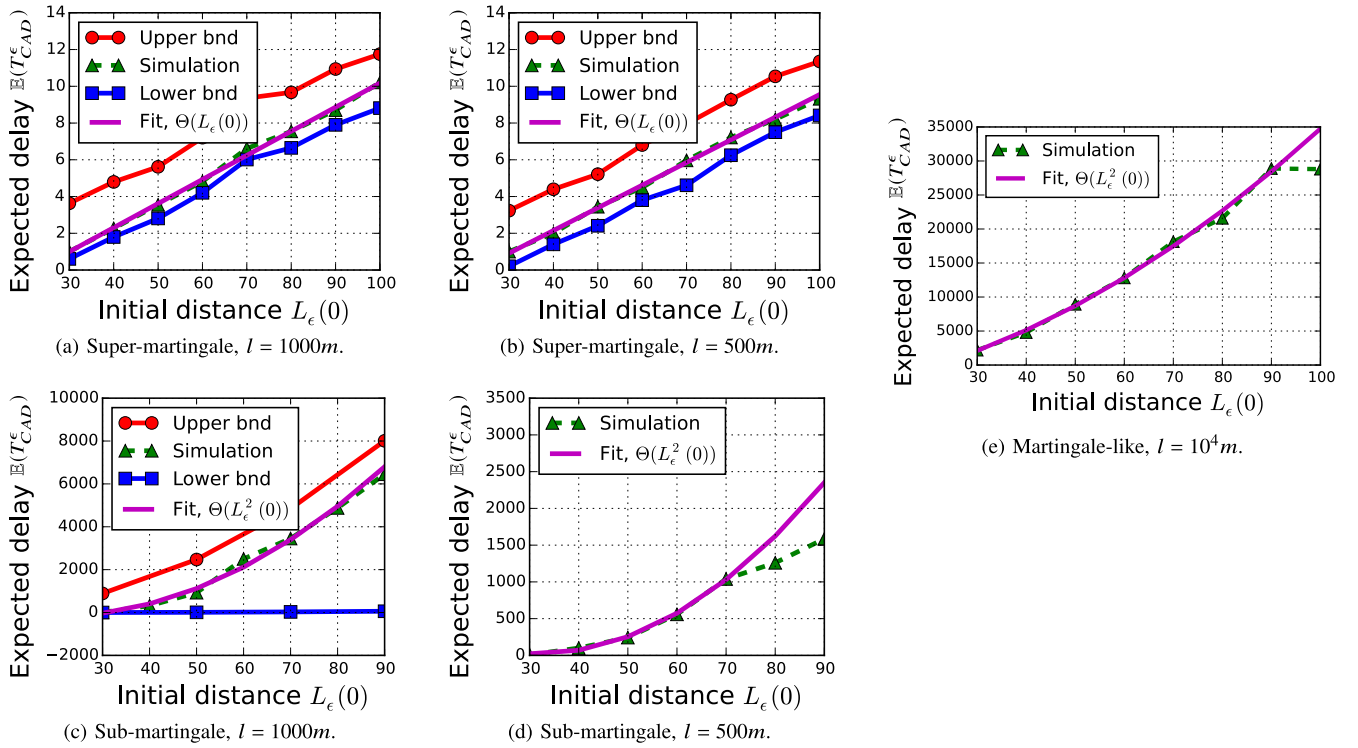


Fig. 5. Expected CAD $\mathbb{E}(T_{CAD}^\epsilon)$ for (a)(b) sub-martingale-like, (c)(d) super-martingale-like, and (e) martingale-like packet mobilities under the boundary effect.

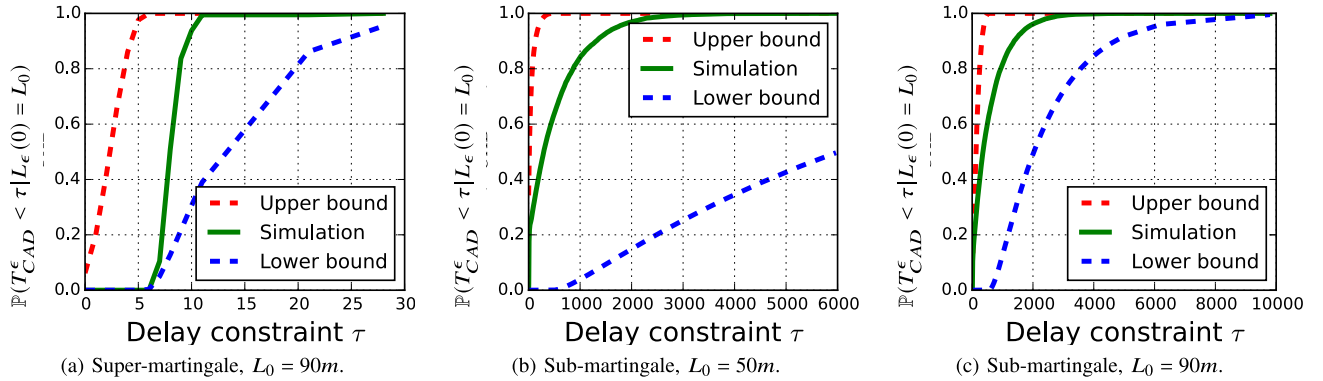


Fig. 6. Theorem 4 holds for SAR $R_{L_0}^\tau$ in both super- and sub-martingale-like packet mobilities.

HOSTILE ENVIRONMENT

Knowing the behavior of the expected CAD against the initial distance-to-destination, and the SAR with respect to certain delay bounds, we move on to the second research question, that is, energy-efficient data delivery to the cloudlet in challenging environments. In these cases, the hand-held devices usually face a tight energy budget. To tackle this practical challenge of stringent delay and tight energy budget, *packet shedding scheme* is proposed to minimize the per-delivery power draw, while maintaining a satisfying SAR. Moreover, a case study is presented to understand the efficacy of the proposed scheme in realistic challenging environments.

A. Toward Energy-Efficient Packet Delivery

A direct implication is that CAD can be greatly reduced by minimizing $\bar{v}_\epsilon$, which may take negative values in super-martingale-like packet mobilities. In line with this obser-

vation, the most widely adopted measure to minimize $\bar{v}_\epsilon$ is incorporating geographical location information in forwarding, such as the motion vector (MoVe) [21] scheme, in which the packet is always forwarded to the α -node closest to destinations. However, in energy-constrained scenarios, it is infeasible to require every α -node on the route to perform such sophisticated forwarding algorithms - some α -nodes may not even have GPS functions due to cost or energy concerns. Therefore, from a practical perspective, we aim to design a simple mechanism to minimize the energy consumption in packet delivery, *i.e.*, per-delivery power draw P_ϵ in Eq. (4).

Unfortunately, minimizing P_ϵ contradicts with the other goal of fast delivery, *i.e.*, minimizing $\mathbb{E}(T_{CAD}^\epsilon)$. For instance, the epidemic routing scheme (R3) achieves a lower CAD thanks to its super-martingale-like type of packet mobility, but it also consumes more energy due to the multi-copy forwarding nature. To strike a balance between a satisfying SAR (for emergency application) and the energy efficiency

for power-constrained end devices in challenging environment, our results on CAD and SAR can be leveraged upon/in packet forwarding schemes. Specifically, packet mobility metrics, such as the overall average speed or SAR, can be directly adopted as the metric to select forwarding targets, executing on every device along the delivery route, *i.e.*, “in” forwarding schemes; or these metrics can be used merely by the data generator to drop futile packets early, which operates “upon” the forwarding scheme, and hence does not affect the delivery routes of data packets.

1) *Packet Shedding Algorithm*: The packet shedding algorithm runs on the packet generating device, which may operate upon any packet forwarding scheme. In this algorithm, a packet ϵ is discarded at time t_ϵ when it is generated, if (according to Theorem 4) the estimated probability $\hat{\mathbb{P}}(T_{CAD}^\epsilon < \tau)$ of a successful delivery is less than a predetermined threshold R_{th} . These early-drop operations are referred to as a *packet shedding* procedure, which runs on every α -node in a distributed manner, as outlined in Algorithm 1.

Algorithm 1 Packet Shedding Procedure

Input: $\mathcal{I} = \{(i, z_i, L_i(z_i), T_{CAD}^i)\}_{i \in I}$, $\mathcal{Q} = \{(q, \tau_q)\}_{q \in \mathcal{Q}}$, r , v_{max} , R_{th}

Output: updated packet queue $\mathcal{Q}$

- 1: **while** $\mathcal{Q} \neq \emptyset$ **do**
 - 2: Take packet ϵ from the queue $(\epsilon, \tau) \leftarrow \mathcal{Q}.\text{pop}()$
 - 3: Record the initial condition $L_\epsilon(0) \leftarrow \text{estimateDist}()$
 - 4: Estimate packet mobility parameters:

$$\hat{v}_\epsilon \leftarrow \frac{1}{|I|} \sum_{i \in I} \frac{r - L_i(z_i)}{T_{CAD}^i}, \hat{\sigma}_\epsilon^2 \leftarrow \frac{1}{|I|} \sum_{i \in I} \left(\frac{L_i(z_i)}{T_{CAD}^i} \right)^2 - \hat{v}_\epsilon^2$$
 - 5:
$$\hat{\mathbb{P}} \leftarrow 1 - \exp \left\{ - \frac{([r - L_\epsilon(0) - \tau \hat{v}_\epsilon]^+)^2}{2(\tau(\hat{\sigma}_\epsilon^2 + (\hat{v}_\epsilon)^2) + \frac{r+v_{max}}{3}[r - L_\epsilon(0) - \tau \hat{v}_\epsilon]^+)} \right\}$$
 - 6: **if** $\hat{\mathbb{P}} < R_{th}$ **then** $\epsilon.\text{transmit}()$
 - 7: **else** $\epsilon.\text{discard}()$
-

The packet shedding procedure does not require GPS function on α -nodes, which is a (power-hungry) prerequisite in many geo-based routing schemes. Instead, our scheme only needs distance information $\{L_i(z_i) | z_i < 0\}_{i \in I} \cup \{L_\epsilon(0)\}$ and CADs $\{T_{CAD}^i\}_{i \in I}$, where I is the index set of successfully delivered packets generated by the current α -node before packet ϵ , and z_i is the generation time of the i -th packet. Specifically, the β -node will broadcast an ACK message whenever a packet ϵ is received, which will be used by α -nodes for three purposes: i) data generating α -nodes can estimate the initial distance $L_i(z_i)$ with the received signal strength or connectivity-based approach widely used in WSNs [32], which is denoted as `estimateDist()` function in the shedding procedure; ii) the generator of this packet ϵ can record T_{CAD}^ϵ in the successful set $\mathcal{I}$; iii) carriers of packet ϵ know it is now safe to discard their copies of ϵ .

The main part of the packet shedding procedure is to estimate the SAR $\hat{\mathbb{P}}$ of a packet ϵ from the packet queue $\mathcal{Q}$, which is later compared with threshold R_{th} determined by specific emergency applications. For this purpose, the overall average speed $\hat{v}_\epsilon$ and variance $\hat{\sigma}_\epsilon^2$ at the moment are first

calculated by the α -node.⁴ Then $\hat{\mathbb{P}}$ is taken as the lower bound of $\mathbb{P}(T_{CAD}^\epsilon < \tau)$ in Eq. (4), to guarantee a satisfying SAR for the worst-case scenario.

Note that packet shedding is performed by α -nodes only at the generation of a packet, but not to packets duplicated/forwarded from nearby α -nodes. The reason behind this design is that though performing the shedding procedure more aggressively may further reduce the power draw, but it also requires every α -node to constantly probe/sense its current location and moving speed of others, potentially consuming more energy during this process. Consequently, the proposed shedding algorithm is transparent to other packets being forwarded in the cloudlet access network.

With respect to computation complexity, both the time complexity and space complexity of the proposed algorithm are low ($O(|\mathcal{Q}|)$ w.r.t. input $\mathcal{I}$ and $\mathcal{Q}$). For time complexity, the entire procedure only contains $O(1)$ algebraic calculations: In step 3, distance estimation can be implemented via direct calculation, or table look-up, both of which are $O(1)$. In step 4, packet mobility parameters are running means and variances, for which Welford’s algorithm can be adopted with $O(1)$ complexity. Step 5 calculates an exponential function of a real number, which is $O(1)$ with respect to the input, as a fixed accuracy (number of digits in $\hat{\mathbb{P}}$) is sufficient. For space complexity, Welford’s algorithm in step 4 is an online algorithm, while the all rest of the steps take no more than $O(|\mathcal{Q}|)$ space to execute.

2) *Forwarding Target Selection Based on Packet Mobility Metrics*: Alternatively, the (current) estimated overall average speed $\bar{v}_\epsilon(t)$ and predicted SAR can be used directly to select the next-hop target whenever the packet is forwarded, substituting the distance-to-destination or node speed metric in GPS-assisted schemes, such as DTN_{geo} in [20]. When incorporated, the packet mobility parameters can be estimated by averaging over historical deliveries, as shown in line 4 of Algorithm 1, whose computation complexity is $O(1)$.

B. Numerical Evaluation

To validate the efficacy of the proposed packet shedding algorithm and the hybrid forwarding schemes in cloudlet access scenarios, we re-employ the configuration in Sec. V-C, except for cell side-length $l = 750m$. When packet shedding is employed, we fix the buffer size on each α -node to $|\mathcal{I}| = 100$. The delay bound τ is set to 5 minutes (300s), which is about the same as the first-response time upon an emergency [24]; and the target SAR R_{sus}^τ is set to 0.8. All the presented results in Fig. 7 are averaged over 3×10^4 network realizations. A k -copy MoVe [21] scheme is taken as the baseline case. MoVe was the first to employ moving direction prediction when selecting forwarding candidates, leading to a negative average packet propagation speed $\bar{v}_\epsilon < 0$, which belongs to the super-martingale-like type of packet mobility, and hence achieves a relatively low CAD in the delay tolerant setting. However, frequent GPS positioning is known as the major

⁴To jump-start the system upon emergency, α -nodes may be required to collect this information in normal conditions. Considering the relatively low overhead to record CAD, this is bearable for most devices.

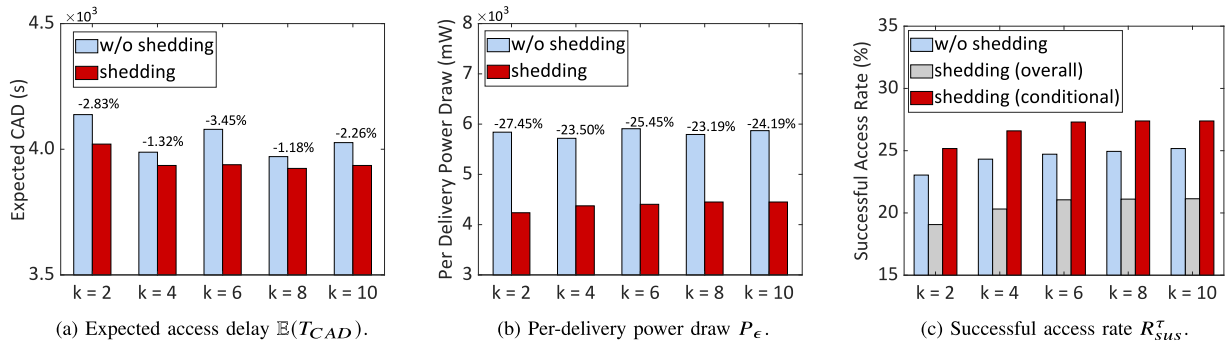


Fig. 7. The proposed packet shedding scheme (on top of a k -copy MoVe forwarding scheme) reduces the power draw by over 20% while the overall SAR is dropped by a mere 4%.

energy drain for many devices. So it is meaningful to examine whether packet shedding can help such schemes save energy, despite the fact that the proposed packet shedding procedure does not require GPS function, and can be applied on top of other packet forwarding schemes. We highlight four observations from the numerical results in Fig. 7.

i) Packets not discarded by the shedding algorithm generally results in a small CAD, as illustrated by Fig. 7a when packet shedding is adopted (red bar). In other words, the proposed algorithm produces a screening effect on packets, that is, packets that are likely to exceed the delay bound τ are filtered out early. It further indicates that the SAR of packet ϵ can be well bounded by Theorem 4, such that estimating it ($\hat{\mathbb{P}}$) with the lower bound in Eq. (9) is an applicable way to determine whether a data transmission is futile to initiate.

ii) Packet shedding saves significantly amount of energy spent on data transmission, even when the number of copies (k) allowed in forwarding is large, as shown in Fig. 7b. This reduction of energy is especially significant when k is small, reaching a staggering 27.45% decrement at $k = 2$. Interestingly, the average P_ϵ is almost constant without shedding, *i.e.*, $\mathbb{E}(P_\epsilon) \approx 5.9$ W irrespective of k . The reason behind this is the trade-off between energy consumption and CAD, *i.e.*, the more copies are allowed in the forwarding scheme, the lower the CAD – in this sense, epidemic routing is the extreme case that forwards to every α -nodes its carrier encounters, and hence achieves the lowest CAD. When packet shedding is employed, the aforementioned trade-off is less visible, and P_ϵ increases with k due to the screening effect.

iii) Packet shedding increases the conditional SAR, which is the number of successfully delivered packets (within delay bound τ) over the total number of *transmitted* packets. As shown in Fig. 7c, shedding increases the conditional SAR by 2.33% on average, indicating that the shedding algorithm selected the more promising packets to forward. Conditional SAR increases with k in both cases, because different copies of a packet can reach the cloudlet along different paths, adding to the likelihood of successful delivery.

iv) The overall SAR R_{sus}^τ , which is defined as the number of successfully delivered packets over the total number of *generated* packets, drops 3.89% percent on average when packet shedding is on. Since $\hat{\mathbb{P}}$ is a first-order estimator, the non-trivial estimation error and the stochastic nature of the packet delivery process leads to this drop, as a necessary cost of energy conservation. Compared with the over 20% energy

saved, this drop of less than 4% is tolerable, especially when data packets are generated at a high frequency. In fact, any α -node may choose its own stance along the delivery-v.s.-power trade-off by adjusting threshold R_{th} to maintain a favorable level of harshness in screening. In this sense, parameter R_{th} and the number of copies k become tunable knobs for α -nodes to adapt to different emergency applications.

To further validate impacts of the defined packet mobility metrics on cloudlet access, we choose the overall average speed to be the virtual link weight in the DTN_{geo} forwarding scheme [20] (square markers, labelled ‘Proposed’), and compared it with the original scheme (triangle markers, labelled ‘GPS-assisted’) and pure epidemic routing (dashed line, labelled ‘Epidemic’) in Fig. 8. Under similar simulation configurations, we observe that incorporating packet mobility characteristics into existing forwarding schemes can greatly reduce the cloudlet access delay in challenging environments, in both the ensemble-mean sense (Fig. 9c and Fig. 9d) and the SAR sense (Fig. 8c). With a comparable CAD performance as the energy-consuming epidemic routing scheme, the average number of delivered packet copies under the proposed scheme (around 5) is much less than the epidemic scheme (more than 10), striking a delicate balance between SAR and energy efficiency for challenging environments.

C. Case Study: Hostile Environment With Packet Loss

To fully understand the characteristics of the packet shedding scheme in a more realistic setting, we consider a cloudlet access scenario with continuous data generation, transmission errors, packet losses (due to noise or interferences), and finer grains of simulation. Basic settings are the same as the previous subsection VI-B, except for the following:

For application, constant bit rate (CBR) traffic is adopted. Specifically, for each simulation instance, 10 out of the 100 α -nodes are randomly chosen as the data generators, who sends out a packet of 512 Bytes every second. Each packet is indexed with a unique sequence number, to examine the impact of the packet shedding scheme on simultaneous data streams.

For routing/forwarding, k -copy MoVe ($k = \{5, 10\}$) is adopted, where neighbor and mobility information is exchanged via short HELLO packets every 0.2 s. Data-carrying α -nodes will try to forward a DATA packet every 1.5 s, or each time a new neighbor is discovered. Expired DATA packets ($\tau = 300$ s) en-route are discarded by the carriers immediately.

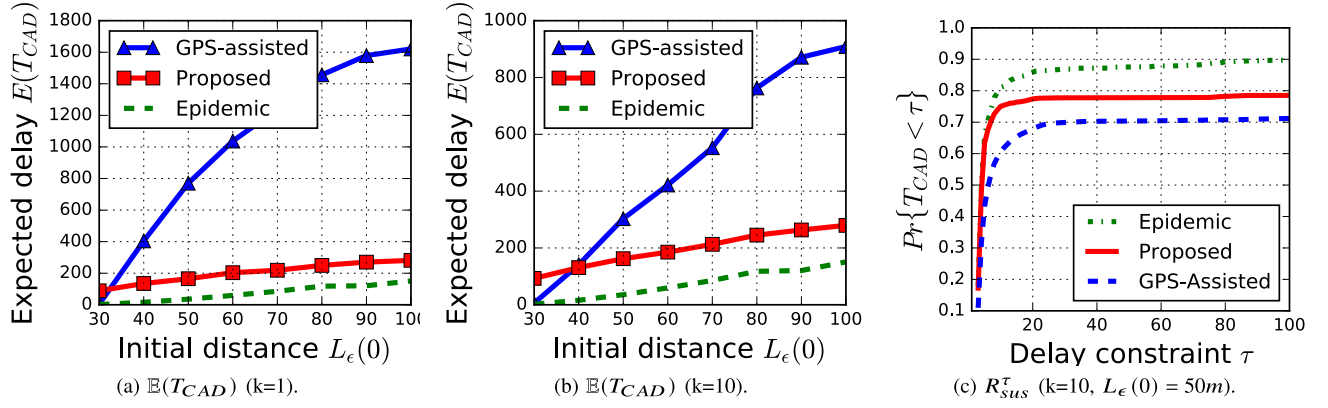


Fig. 8. Adopting the *overall average speed* in the DTN_{geo} forwarding scheme greatly reduces the expected CAD, pushing the delay performance close to the epidemic routing scheme, which consumes the most energy due to its unlimited number of forwarded packet copies.

TABLE III
PERFORMANCE STATISTICS

Schemes	$k = 5, \text{PER} = 0.01$		$k = 10, \text{PER} = 0.01$	
	MoVe	MoVe-Shedding	MoVe	Move-Shedding
# of Deliveries (Unique/Total)	539 / 1452 (37.12%)	458 / 986 (46.45%)	754 / 1798 (41.94%)	788 / 1883 (41.85%)
SAR (%)	5.39%	4.58%	7.54%	7.88%
Avg. CAD (s)	60.49 / 91.08	50.02 / 70.89	82.77 / 106.1	99.65 / 118.92
# of Forwards per Delivery	267.85 / 99.4	237.44 / 110.22	379.41 / 159.11	299.78 / 128.88
Avg. # of Successful Accesses	5.3	5.1	6.4	7.1

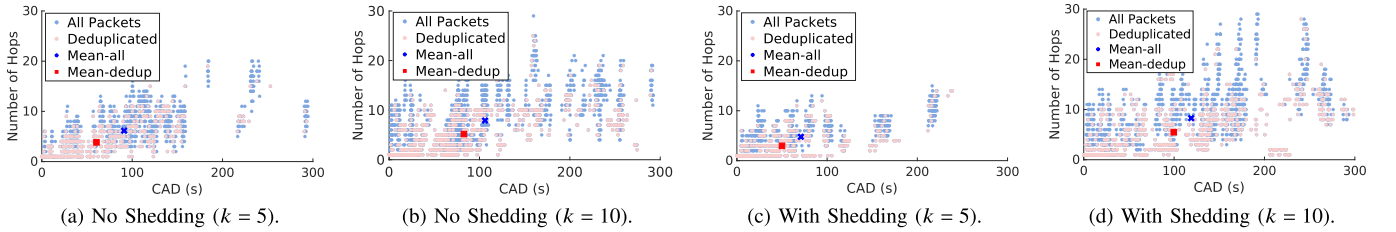


Fig. 9. Cloudlet Access Delay (CAD) and hop counts of the delivered data packets.

For communication, each α -node is equipped with a radio of range 25 m, transmission power 2 mW, and transmission rate of 2 Mbps. The link layer protocol is pure ALOHA, so collision may result in packet retransmissions. Each packet (including both HELLO and DATA) is subject to a 0.01 loss rate during transmissions, to simulate the noise and interference in hostile environments.

All α -nodes follows RWP (T2) mobility, with a maximum speed of 10 m/s, pausing time [3, 8] s, and walking time [10, 30] s, followed from human walking patterns. Each simulation instance is run for 1000 s, and events are updated every discrete time steps of length 100 ms.

Performance statistics of both the vanilla MoVe and MoVe-Shedding are presented in Table III. It is worth noticing that the proposed packet shedding scheme i) greatly reduces the number of forwarded copies during the cloudlet access process, achieving the energy-conserving goal in challenging environments; ii) exhibits different SAR and CAD behavior under $k = 5$ and $k = 10$ settings. Comparing with the $k = 5$ case, the shedding algorithm obtains more performance gain in the large k scenario, manifested by the higher SAR, the higher number of (uniquely) delivered data packets, and the larger number of data generators with successful cloudlet access. To fully examine how shedding improves the efficacy of cloudlet access, we plot the CAD v.s. hop-counts statistics of successful deliveries in Figure 9. For $k = 10$, the slightly

higher average hop-count indicates that shedding allows more slow packets to “barely access” the cloudlet within the delay bound, as illustrated by denser light-red dots to the right of Fig. 9(d). This effect is not present in the case of $k = 5$, however, indicating that a variable shedding threshold R_{sus}^τ could be adopted to further improve the efficacy of cloudlet access in hostile environments.

VII. CONCLUSION

We study the problem of delivering data packets to infrastructure cloudlets in challenging environments, whose performance can be characterized by the cloudlet access delay (CAD), energy consumption per packet delivery, and the successful access rate (SAR). To model the relative movements of packets toward the cloudlet, we introduce the notion of packet mobility, which can be categorized into sub-martingale, martingale, and super-martingale-like types based on the average propagation speed of packets. Our analysis and simulation results revealed the scaling behavior of CAD with respect to different packet mobility types, which can be leveraged to estimate QoS of cloudlet-mounted services, and design applications for various scenarios. As an exempling application, we develop a distributed, low-complexity packet shedding scheme to reduce the power consumption of packet delivery in challenging environments, and its migration with existing forwarding schemes. We believe the proposed analysis

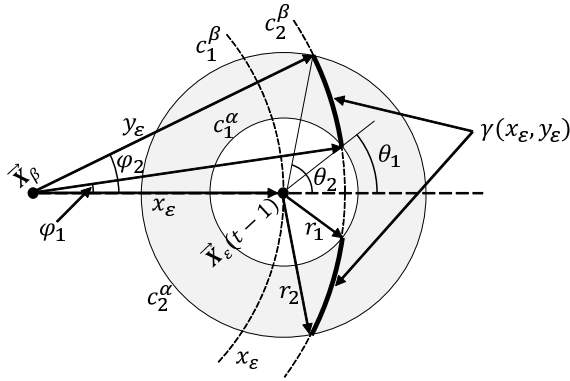


Fig. 10. Example of the geometric approach.

framework can be further extended to study the impact of device density, as well as grouped mobility on CAD and SAR performances, to achieve a comprehensive understanding of the cloudlet access problem. In a more practical sense, content-related and semantic factors of packets can be considered together with the SAR in the shedding scheme, such that packet delivery services can be fine-tuned to provision for diversified cloudlet access applications.

APPENDIX A PROOF OF LEMMA 1

By the definition of a (sub-/super-)martingale [29], we need to check against three criteria.

- [C1] $L_\epsilon(t)$ is $\mathcal{F}_t$ -measurable, where $\mathcal{F}_t$ is a filtration on probability space $(\Omega, \mathcal{F}, \mathbb{P})$: Consider natural filtration $\mathcal{F}_t = \sigma(v_\epsilon(0), v_\epsilon(1), \dots, v_\epsilon(t))$, where $\sigma(\mathcal{U})$ is the smallest σ -algebra containing set $\mathcal{U} \subset \Omega$. Since $L_\epsilon(t) = L_\epsilon(0) + \sum_{k=1}^t v_\epsilon(k)$ is a function of $v_\epsilon(0), v_\epsilon(1), \dots, v_\epsilon(t)$, process $L_\epsilon(t)$ is $\mathcal{F}_t$ -measurable.
- [C2] $\mathbb{E}(|L_\epsilon(t)|) < \infty$ for all $t \in \mathcal{T}$: We have $\mathbb{E}(|L_\epsilon(t)|) = \mathbb{E}(|L_\epsilon(0) + \sum_{k=1}^t v_\epsilon(k)|) \leq \mathbb{E}(|L_\epsilon(0)|) + t\mathbb{E}(|v_\epsilon(k)|)$. As $|L_\epsilon(0)|$ is finite, and $v_\epsilon(k)$ is bounded, so $\mathbb{E}(|L_\epsilon(t)|) < \infty$.
- [C3] $\mathbb{E}(L_\epsilon(t)|\mathcal{F}_{t-1}) = L_\epsilon(t-1)$ for martingales: By definition, $\mathbb{E}(L_\epsilon(t)|\mathcal{F}_{t-1}) = \mathbb{E}(L_\epsilon(t-1) + v_\epsilon(t)|\mathcal{F}_{t-1}) = L_\epsilon(t-1) + \mathbb{E}(v_\epsilon(t)|\mathcal{F}_{t-1}) = L_\epsilon(t-1)$.

Note that the equal sign in condition C3 holds for the case of $\mathbb{E}(v_\epsilon(t)) = 0$. When $\mathbb{E}(v_\epsilon(t)) < 0$ (respectively > 0), process $\{L_\epsilon(t)\}_{t \in \mathcal{T}}$ is a sub-martingale (respectively super-martingale).

APPENDIX B PROOF OF COROLLARY 1 AND 2

To show Corollary 1, i.e., $\mathbb{E}(L_\epsilon(t)|\mathcal{F}_{t-1}) > L_\epsilon(t-1)$, we take a geometric approach similar to that in [18], while our focus is the distance-to-destination $L_\epsilon(t)$. Since direct contact routing (R1) is used, the movements of packet ϵ are mainly due to the random walk node mobility (T1).

In T1, a packet-carrying α_i -node moves from its previous location $\vec{X}_\epsilon(t-1) = \{\vec{X}_{\alpha_i}(t-1)\}$ to a new location by randomly choosing a speed v_{node} and a direction θ uniformly at random from ranges $[v_{min}, v_{max}]$ and $[0, 2\pi]$, respectively.

Thus, the location of node α_i at the next time step, $\vec{X}_\epsilon(t) = \{\vec{X}_{\alpha_i}(t)\}$, is within ring area $\{c_2^\alpha - c_1^\alpha\}$ where c_2^α and c_1^α are the arcs centered at $\vec{X}_\epsilon(t-1)$ with radii $r_2 = v_{max}$ and $r_1 = v_{min}$, respectively, as depicted in Fig. 10. Furthermore, let $\mathbb{P}\{y_\epsilon|x_\epsilon\} := \mathbb{P}\{L_\epsilon(t) = y_\epsilon | L_\epsilon(t-1) = x_\epsilon\}$ be the progress probability that node α_i located at $\vec{X}_\epsilon(t-1)$ with a distance-to-destination x_ϵ moves to a new location $\vec{X}_\epsilon(t)$ with a distance-to-destination of y_ϵ . Then,

$$\begin{aligned} \mathbb{E}(L_\epsilon(t)|\mathcal{F}_{t-1}) &= \mathbb{E}(L_\epsilon(t)|L_\epsilon(t-1) = x_\epsilon) \\ &= \int_{x_\epsilon - r_2}^{x_\epsilon + r_2} y_\epsilon \mathbb{P}\{y_\epsilon|x_\epsilon\} dy_\epsilon. \end{aligned} \quad (11)$$

To calculate $\mathbb{P}\{y_\epsilon|x_\epsilon\}$, let c_2^β be an arc with radius y_ϵ centered at the β -node's location $\vec{X}_\beta$. Let $\gamma(x_\epsilon, y_\epsilon)$ be the length of the intersection between the arc c_2^β with the ring area $\{c_2^\alpha - c_1^\alpha\}$, as shown in Fig. 10. Then, it is clear that $\mathbb{P}\{y_\epsilon|x_\epsilon\} = \frac{\varphi x_\epsilon}{\pi(r_2^2 - r_1^2)}$, where

$$\varphi = \begin{cases} \varphi_2 - \varphi_1, & \text{if } |y_\epsilon - x_\epsilon| \leq r_1, \\ \varphi_2, & \text{if } r_1 < |y_\epsilon - x_\epsilon| \leq r_2; \end{cases} \quad (12)$$

φ_2 is the angle formed by the the point $\vec{X}_\epsilon(t-1)$, the center of the cell $\vec{X}_\beta$, and the outermost point of arc $\gamma(x_\epsilon, y_\epsilon)$; and φ_1 is the angle formed by the two former points and the innermost point of arc $\gamma(x_\epsilon, y_\epsilon)$ after removing the area $c(\vec{X}_\epsilon(t-1), r_1)$. Here, "innermost" refers to a point of an arc closest to the line connecting the center point $\vec{X}_\beta$ and $\vec{X}_\epsilon(t-1)$, while $c(\vec{X}, r)$ denotes a solid circle centered at $\vec{X}$ with radius r .

Let θ_1 and θ_2 be defined as the two new angles depicted in Fig. 10, then we can obtain

$$y_\epsilon^2 = (x_\epsilon + r_1 \cdot \cos(\theta_1))^2 + (r_1 \cdot \sin(\theta_1))^2, \quad (13)$$

$$y_\epsilon \cdot \cos(\varphi_1) = x_\epsilon + r_1 \cdot \cos(\theta_1). \quad (14)$$

Substituting Eq. (14) into Eq. (13) results into a closed-form solution $\varphi_1 = \arccos\left\{\frac{x_\epsilon^2 + y_\epsilon^2 - r_1^2}{2x_\epsilon y_\epsilon}\right\}$, and φ_2 can be obtained similarly. Finally, plugging φ_1 and φ_2 into Eq. (12) and into $\mathbb{P}\{y_\epsilon|x_\epsilon\} = \frac{\varphi x_\epsilon}{\pi(r_2^2 - r_1^2)}$, we obtain $\mathbb{E}(L_\epsilon(t)|\mathcal{F}_{t-1}) > x_\epsilon$ after solving the integral in (11) numerically.

For the case of T1 (Infostation) + R2, recall that region $\mathcal{N}$ is a line segment, the two-way traffic has identical node speed distribution, and the packet ϵ is forwarded to a randomly-picked contact. Consequently, the progress probability $\mathbb{P}\{L_\epsilon(t) = y_\epsilon | L_\epsilon(t-1) = x_\epsilon\}$ follows the uniform distribution with range $[x_\epsilon - (r + v_{node}), x_\epsilon + (r + v_{node})]$. Plugging such $\mathbb{P}\{y_\epsilon|x_\epsilon\}$ into Eq. (11) yields $\mathbb{E}(L_\epsilon(t)|\mathcal{F}_{t-1}) = x_\epsilon = L_\epsilon(t-1)$, indicating a martingale type of packet mobility.

APPENDIX C PROOF OF THEOREM 1

Define $L'_\epsilon(t) := L_\epsilon(t) - H_t$, where $H_t := \sum_{k=1}^t \mathbb{E}(v_\epsilon(k)|\mathcal{F}_{k-1})$. When $\gamma \rightarrow 0$, $\{L_\epsilon(t)\}_{t \in \mathcal{T}}$ becomes a supermartingale and we have $\mathbb{E}(L'_\epsilon(t)|\mathcal{F}_{t-1}) = \mathbb{E}(L_\epsilon(t) - v_\epsilon(t) - H_{t-1} | \mathcal{F}_{t-1}) = L_\epsilon(t-1) - H_{t-1} = L'_\epsilon(t-1)$, which implies that the process $\{L'_\epsilon(t)\}_{t \in \mathcal{T}}$ is a martingale.

Notice that T_{CAD}^ϵ is a *stopping time* [29], [33] since it depends on $\vec{X}_\epsilon(t)$ up to the current time t (not the future), and is $\mathcal{F}_t$ -measurable. To analyze the stopping time with Doob's optional stopping theorem (OST) [29], as that in the *Gambler's ruin* problem [33] and the *Random walk* problem, we need to check whether T_{CAD} is finite.

Observe that for time step $t \in \mathcal{T}$, $\{v_\epsilon(t)\}_{t \in \mathcal{T}}$ are *independent* r.v.'s. The reason behind this observation is that when the packet's location at the current time t —and the corresponding $L_\epsilon(t)$ —is independent to the locations at the previous time steps (captured by $\{L_\epsilon(k)\}_{k < t}$, e.g., under RW node mobility without routing. On the other hand, when routing is employed under a more general node mobility model, packet ϵ is constantly copied or forwarded to the newly-contacted nodes. The positions of these contacted nodes are known to be uncorrelated to the locations of the packet-carrying α -nodes from the previous time steps [34, Sec. 2]. Thus, the independent observation is also valid when routing is employed, since there is a high chance of contacting a new α -node during every time step, especially as the number of nodes n increases.

Let $T' := \inf\{t : L_\epsilon(0) + \sum_{k=1}^t v_k^\epsilon = 0\}$ be the time until a packet ϵ hits the origin point, $\vec{X}_\beta$. Clearly, $T_{CAD}^\epsilon < T'$. Since $\{v_\epsilon(t)\}_{t \in \mathcal{T}}$ are i.i.d., Wald's first equality implies that $\mathbb{E}(T_{CAD}^\epsilon) < \mathbb{E}(T') = \frac{L_\epsilon(0)}{-\mathbb{E}(v_\epsilon(t)|\mathcal{F}_{t-1})} < \infty$. This inequality, combined with the fact that $\mathbb{E}(|L'_\epsilon(t) - L'_\epsilon(t-1)||\mathcal{F}_{t-1}) = \mathbb{E}(L'_\epsilon(t) - L'_\epsilon(t-1)|\mathcal{F}_{t-1}) = 0$, indicate that the two conditions for applying the OST [33] are satisfied. Accordingly, since $\mathbb{E}(L'_\epsilon(T_{CAD}^\epsilon)|\mathcal{F}_0) = L'_\epsilon(0)$,

$$\begin{aligned} \mathbb{E}((L_\epsilon(T_{CAD}) - H_{T_{CAD}^\epsilon}^\epsilon) - (L_\epsilon(0) - H_0)) &= 0 \\ \Leftrightarrow \mathbb{E}(L_\epsilon(0) - L_\epsilon(T_{CAD})) &= -H_{T_{CAD}^\epsilon}^\epsilon \end{aligned} \quad (15)$$

Upon delivery, packet ϵ is already within the transmission range of the β -node, that is, $0 \leq L_\epsilon(T_{CAD}) \leq r$, so $L_\epsilon(0) - r \leq -H_{T_{CAD}^\epsilon}^\epsilon \leq L_\epsilon(0)$. Also, $\bar{v}_\epsilon = \mathbb{E}_{\mathcal{F}_{t-1}}(\mathbb{E}(v_\epsilon(t)|\mathcal{F}_{t-1}))$ and $\mathbb{E}(H_{T_{CAD}^\epsilon}^\epsilon) = \bar{v}_\epsilon \mathbb{E}(T_{CAD}^\epsilon)$ by Wald's equation. Plugging this result into (15) obtains Theorem 1.

APPENDIX D PROOF OF THEOREM 2

Consider the time instant $t = T_{CAD}^\epsilon$ when packet ϵ is delivered. Since $L_\epsilon(t) = L_\epsilon(0) + \sum_{i=1}^t v_\epsilon(i)$, applying the expectation operator $\mathbb{E}(\cdot)$ to both sides of Eq. (7), we obtain

$$\begin{aligned} \sum_{i=1}^{T_{CAD}^\epsilon} \mathbb{E}(v_\epsilon(i)^2|\mathcal{F}_{i-1}) \\ = \mathbb{E}((L_\epsilon(T_{CAD}^\epsilon) - L_\epsilon(0))^2) \\ - 2 \sum_{i=2}^{T_{CAD}^\epsilon} \mathbb{E}(v_\epsilon(i)|\mathcal{F}_{i-1}) (L_\epsilon(i-1) - L_\epsilon(0)). \end{aligned} \quad (16)$$

For sub-martingale like packet mobilities, $\mathbb{E}(v_\epsilon(i)|\mathcal{F}_{i-1}) \geq 0$ and $L_\epsilon(0) \geq 0$ for all $t \in \mathcal{T}$, so

$$\begin{aligned} \mathbb{E}((L_\epsilon(0) - L_\epsilon(T_{CAD}^\epsilon))^2) \\ \stackrel{Wald}{\leq} \sum_{i=1}^{T_{CAD}^\epsilon} [\mathbb{E}(v_\epsilon(i)^2|\mathcal{F}_{i-1}) + 2(v_{max} + r)\mathbb{E}(v_\epsilon(i)|\mathcal{F}_{i-1})] \\ = \mathbb{E}(T_{CAD}^\epsilon) [\mathbb{E}(v_{T_{CAD}^\epsilon}^2|\mathcal{F}_{t-1}) + 2(v_{max} + r)\mathbb{E}(v_{T_{CAD}^\epsilon}|\mathcal{F}_{t-1})] \\ = \mathbb{E}(T_{CAD}^\epsilon) [\bar{\sigma}^2 + (\bar{v}_\epsilon)^2 + 2(v_{max} + r)\bar{v}_\epsilon]. \end{aligned} \quad (17)$$

Moreover, $L_\epsilon(T_{CAD}^\epsilon) \leq r$ and $\mathbb{E}(L_\epsilon(0)) = L_\epsilon(0)$, hence we have proved the left hand side of Eq. (7). For the right hand side, we re-state (16) as follows.

$$\begin{aligned} \mathbb{E}((L_\epsilon(0) - L_\epsilon(T_{CAD}^\epsilon))^2) \\ \geq \mathbb{E}(T_{CAD}^\epsilon) \cdot \mathbb{E}(v_\epsilon(t)^2|\mathcal{F}_{t-1}) \\ = \mathbb{E}(T_{CAD}^\epsilon) [\bar{\sigma}^2 + (\bar{v}_\epsilon)^2]. \end{aligned} \quad (18)$$

Finally, the upper bound in Eq. (7) can be obtained by plugging $L_\epsilon(T_{CAD}^\epsilon) \geq 0$ into Eq. (18).

APPENDIX E PROOF OF THEOREM 3

Define auxiliary process $\{Y_t\}_{t \in \mathcal{T}}$, where $Y_t := (L_\epsilon(0) - L_\epsilon(t))^2 - S_t$ and $S_t := \sum_{k=1}^t \sigma_k^2$. Then for a martingale-like packet mobility when $\gamma = b_i - a_i \rightarrow 0$ ($a_i \leq L_\epsilon(t-1) < b_i$), we have

$$\begin{aligned} \mathbb{E}(Y_t|\mathcal{F}_{t-1}) &= \mathbb{E}([(L_\epsilon(0) - L_\epsilon(t-1))^2 - S_{t-1} \\ &\quad - 2v_t(L_\epsilon(0) - L_\epsilon(t-1)) + v_\epsilon(t)^2 - \sigma_t^2]|\mathcal{F}_{t-1}). \\ &= \mathbb{E}([(L_\epsilon(0) - L_\epsilon(t-1))^2 - S_{t-1}]|\mathcal{F}_{t-1}) = Y_{t-1}, \end{aligned} \quad (19)$$

because $\mathbb{E}(L_\epsilon(0) - L_\epsilon(t-1)|\mathcal{F}_{t-1}) = 0$ and $\sigma_t^2 = \mathbb{E}(v_\epsilon(t)^2)$. Eq. (19) implies that $\{Y_t\}_{t \in \mathcal{T}}$ is also a martingale. To apply OST as in the proof of Theorem 1, we need to check the following conditions: i) $\mathbb{E}(T_{CAD}^\epsilon) < \infty$, and ii) $\mathbb{E}(|Y_t - Y_{t-1}||\mathcal{F}_{t-1}) \leq c$ for all $t \in \mathcal{T}$ and some constant c .

For condition i), we re-employ the time to hit the origin, T' , as in Appendix C, and applying Wald's second equality, we have $\mathbb{E}(T_{CAD}^\epsilon) \leq \mathbb{E}(T') = \frac{\mathbb{E}(L_\epsilon(0))^2}{\bar{\sigma}^2} < \infty$.

For condition ii), $|Y_t - Y_{t-1}| = |L_\epsilon^2(t) - 2L_\epsilon(0)(L_\epsilon(t) - L_\epsilon(t-1)) - |L_\epsilon^2(t-1)| - \sigma_t^2|$, so

$$\begin{aligned} \mathbb{E}(|Y_t - Y_{t-1}||\mathcal{F}_{t-1}) \\ = \mathbb{E}(|L_\epsilon^2(t) - 2L_\epsilon(0)(L_\epsilon(t) - L_\epsilon(t-1))| \\ - |L_\epsilon^2(t-1)| - \sigma_t^2|\mathcal{F}_{t-1}) \\ \leq \mathbb{E}(|L_\epsilon^2(t) - L_\epsilon^2(t-1)||\mathcal{F}_{t-1}) \\ \leq (\sqrt{2}l - (r + v_{max}))(r + v_{max}), \end{aligned} \quad (20)$$

because the highest value of $\mathbb{E}(|Y_t - Y_{t-1}||\mathcal{F}_{t-1})$ is when the packet ϵ moves from the corner of the cell towards the β -node with a maximum movement propagation distance of $r + v_{max}$.

Applying Doob's OST to the auxiliary process, we have $\mathbb{E}((L_\epsilon(0) - L_\epsilon(T_{CAD}^\epsilon))^2) = S_{T_{CAD}^\epsilon}^\epsilon$, where $(L_\epsilon(0) - r)^2 \leq S_{T_{CAD}^\epsilon}^\epsilon \leq (L_\epsilon(0))^2$. As $\{v_\epsilon(t)\}$ are i.i.d., $\mathbb{E}(S_{T_{CAD}^\epsilon}^\epsilon) = \mathbb{E}(\sum_{k=1}^{T_{CAD}^\epsilon} \sigma_k^2) = \bar{\sigma}^2 \mathbb{E}(T_{CAD}^\epsilon)$. Plugging it into the inequality above yields Theorem 3.

REFERENCES

- [1] T. Luxner, "Cloud computing trends: 2021 state of the cloud report," Flexera, Mar. 2022. Accessed: Oct. 2, 2022. [Online]. Available: <https://www.flexera.com/blog/cloud/cloud-computing-trends-2022-state-of-the-cloud-report/>
- [2] M. Goudarzi, H. Wu, M. Palaniswami, and R. Buyya, "An application placement technique for concurrent IoT applications in edge and fog computing environments," *IEEE Trans. Mobile Comput.*, vol. 20, no. 4, pp. 1298–1311, Apr. 2021.
- [3] J. He, J. Wei, K. Chen, Z. Tang, Y. Zhou, and Y. Zhang, "Multitier fog computing with large-scale IoT data analytics for smart cities," *IEEE Internet Things J.*, vol. 5, no. 2, pp. 677–686, Apr. 2018.

- [4] T. Q. Thinh, J. Tang, Q. D. La, and T. Q. S. Quek, "Offloading in mobile edge computing: Task allocation and computational frequency scaling," *IEEE Trans. Commun.*, vol. 65, no. 8, pp. 3571–3584, Aug. 2017.
- [5] A. M. Hegland, M. Hauge, and A. Holtzer, "Federating tactical edge networks: Ways to improve connectivity, security, and network efficiency in tactical heterogeneous networks," *IEEE Commun. Mag.*, vol. 58, no. 2, pp. 72–78, Feb. 2020.
- [6] M. Babar, M. S. Khan, F. Ali, M. Imran, and M. Shoaib, "Cloudlet computing: Recent advances, taxonomy, and challenges," *IEEE Access*, vol. 9, pp. 29609–29622, 2021.
- [7] M. Satyanarayanan, G. Lewis, E. Morris, S. Simanta, J. Boleng, and K. Ha, "The role of cloudlets in hostile environments," *IEEE Pervasive Comput.*, vol. 12, no. 4, pp. 40–49, Oct. 2013.
- [8] N. Fernando, S. W. Loke, and W. Rahayu, "Mobile cloud computing: A survey," *Future Gen. Comput. Syst.*, vol. 29, no. 1, pp. 84–106, 2013.
- [9] R. H. Frenkiel, B. R. Badrinath, J. Borres, and R. D. Yates, "The Infostations challenge: Balancing cost and ubiquity in delivering wireless data," *IEEE Pers. Commun.*, vol. 7, no. 2, pp. 66–71, Apr. 2000.
- [10] R. Groenevelt, P. Nain, and G. Koole, "The message delay in mobile ad hoc networks," *Perform. Eval.*, vol. 62, nos. 1–4, pp. 210–228, 2005.
- [11] S. Medjiah, T. Taleb, and T. Ahmed, "Sailing over data mules in delay-tolerant networks," *IEEE Trans. Wireless Commun.*, vol. 13, no. 1, pp. 5–13, Jan. 2014.
- [12] X. Zhang, G. Neglia, J. Kurose, and D. Towsley, "Performance modeling of epidemic routing," *Comput. Netw.*, vol. 51, nos. 10–11, pp. 2867–2891, Jul. 2007.
- [13] K. Mitra, Saguna, and C. Ahlund, "A mobile cloud computing system for emergency management," *IEEE Cloud Comput.*, vol. 1, no. 4, pp. 30–38, Nov. 2014.
- [14] P. Bellavista, A. Zanni, and M. Solimando, "A migration-enhanced edge computing support for mobile devices in hostile environments," in *Proc. 13th Int. Wireless Commun. Mobile Comput. Conf. (IWCMC)*, Jun. 2017, pp. 957–962.
- [15] T. Camp, J. Boleng, and V. Davies, "A survey of mobility models for ad hoc network research," *Wireless Commun. Mobile Comput.*, vol. 2, no. 5, pp. 483–502, Aug. 2002.
- [16] H. Cai and D. Y. Eun, "Crossing over the bounded domain: From exponential to power-law inter-meeting time in MANET," in *Proc. 13th Annu. ACM Int. Conf. Mobile Comput. Netw.*, Sep. 2007, pp. 159–170.
- [17] A. Picu and T. Spyropoulos, "DTN-meteo: Forecasting the performance of DTN protocols under heterogeneous mobility," *IEEE/ACM Trans. Netw.*, vol. 23, no. 2, pp. 587–602, Apr. 2015.
- [18] C.-F. Hsin and M. Liu, "Hitting time analysis for a class of random packet forwarding schemes in ad hoc networks," *Ad Hoc Netw.*, vol. 7, no. 3, pp. 500–513, May 2009.
- [19] M. Asadpour, S. Egli, K. A. Hummel, and D. Giustiniano, "Routing in a fleet of micro aerial vehicles: First experimental insights," in *Proc. 3rd ACM Workshop Airborne Netw. Commun.*, Aug. 2014, pp. 9–10.
- [20] M. Asadpour, K. A. Hummel, D. Giustiniano, and S. Draskovic, "Route or carry: Motion-driven packet forwarding in micro aerial vehicle networks," *IEEE Trans. Mobile Comput.*, vol. 16, no. 3, pp. 843–856, Mar. 2017.
- [21] J. LeBrun, C.-N. Chuah, D. Ghosal, and M. Zhang, "Knowledge-based opportunistic forwarding in vehicular wireless ad hoc networks," in *Proc. IEEE 61st Veh. Technol. Conf.*, May 2005, pp. 2289–2293.
- [22] N. A. Pantazis, S. A. Nikolidakis, and D. D. Vergados, "Energy-efficient routing protocols in wireless sensor networks: A survey," *IEEE Commun. Surveys Tuts.*, vol. 15, no. 2, pp. 551–591, 2nd Quart., 2012.
- [23] T. Spyropoulos, K. Psounis, and C. S. Raghavendra, "Single-copy routing in intermittently connected mobile networks," in *Proc. 1st Annu. IEEE Commun. Soc. Conf. Sensor Ad Hoc Commun. Netw. (SECON)*, Oct. 2004, pp. 235–244.
- [24] A. Ingolfsson, S. Budge, and E. Erkut, "Optimal ambulance location with random delays and travel times," *Health Care Manage. Sci.*, vol. 11, no. 3, pp. 262–274, Sep. 2008.
- [25] I. Rhee et al. (Jul. 2009). *CRAWDAD Dataset Ncsu/Mobilitymodels*. Accessed: Jul. 23, 2009. Downloaded From [Online]. Available: <http://crawdad.org/ncsu/mobilitymodels/20090723>
- [26] T. Winter and P. Thubert, *RPL: IPv6 Routing Protocol for Low-Power and Lossy Networks*, document RFC 6550, RFC Editor, Internet Requests for Comments, Mar. 2012, pp. 1–157. [Online]. Available: <https://www.rfc-editor.org/rfc/rfc6550#page-154>
- [27] L. Gupta, R. Jain, and G. Vaszkun, "Survey of important issues in UAV communication networks," *IEEE Commun. Surveys Tuts.*, vol. 18, no. 2, pp. 1123–1152, 2nd Quart., 2016.
- [28] M. Grossglauser and D. Tse, "Mobility increases the capacity of ad-hoc wireless networks," in *Proc. IEEE INFOCOM*, vol. 3, Apr. 2001, pp. 1360–1369.
- [29] J. Jacod and P. Protter, *Probab. Essentials*. Cham, Switzerland: Springer, 2004.
- [30] R. Shirani, M. St-Hilaire, T. Kunz, Y. Zhou, J. Li, and L. Lamont, "Quadratic estimation of success probability of greedy geographic forwarding in unmanned aeronautical ad-hoc networks," in *Proc. IEEE 75th Veh. Technol. Conf. (VTC Spring)*, May 2012, pp. 1–5.
- [31] F. Chung and L. Lu, "Concentration inequalities and Martingale inequalities: A survey," *Internet Math.*, vol. 3, no. 1, pp. 79–127, 2006.
- [32] Q. Miao, B. Huang, and B. Jia, "Estimating distances via received signal strength and connectivity in wireless sensor networks," *Wireless Netw.*, vol. 26, no. 2, pp. 971–982, Feb. 2020.
- [33] G. Grimmett and D. Stirzaker, *Probability and Random Processes*. London, U.K.: Oxford Univ. Press, 2001.
- [34] F. Bai and A. Helmy, "A survey of mobility models," in *Wireless Adhoc Networks*, vol. 206. Los Angeles, CA, USA: Univ. Southern California, 2004, p. 147.



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